

The ring of character values depends only on the segment: $SU(n)$, $Sp(2m)$ and $GL(n)$ at a torsion element

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Abstract

Let $g_q = \exp(2\pi i \rho/q)$ be the principal element attached to q in a compact simple group, and let A be the ring generated by all character values at g_q , an order in the integers $\mathcal{O}$ of its field. A division lemma — multiplying a monic polynomial by a monic integer factor does not change the ring generated by its coefficients — gives three operations on the segment of exponents that leave A unchanged: adjoining an eigenvalue 1, which makes the ring of $SU(2m+1)$ equal to that of $Sp(2m)$; the complement $n \mapsto q-n$, level-rank duality as an equality of orders; and the period $n \mapsto n+q$. So A depends only on the segment. At p , the index of a stable segment is the sum of its layer defects, and the relative class number governs the first layer. A prime p divides $[\mathcal{O} : A]$ only if the prime-to- p part r of q divides $n-1$, n or $n+1$, and inside these families it does whenever p is an odd prime dividing q and $\varphi(r) \geq 4$. For the non-real ring of $GL(n)$, with $n = Nr$ or $Nr+1$, $2 \leq n < q$, $r \geq 5$ odd and p odd, the order is locally a free quadratic extension of the ring of the recentred segment.

Keywords: orders generated by character values; ring of integers; index of an order; compact Lie groups; elements of finite order; level–rank duality; quantum dimensions; relative class number; cyclotomic fields; Gorenstein orders.

1 Fifty years of values, and a nineteenth-century question

A character value at an element of order q is a sum of q -th roots of unity, and in general that is all one can say. At one element much more can be said, and it has been said three times. In 1976 Kostant proved that at the Coxeter conjugacy class every irreducible character of a compact simple group takes the value 1, 0 or -1 , and that those values *are* the coefficients of Macdonald’s η -function identity [12, Thm. 3.1]: a single conjugacy class, and a classical product formula reads off it. Kostant later determined which weights contribute — those with $\lambda + \rho$ in the orbit of ρ under the affine Weyl group at level $\frac{1}{2}$ [13] — and Cellini, Möseneder Frajria and Papi analysed that orbit and made the description explicit, in the classical types, as a statement about affine Weyl groups permuting the integers [3]. And in 2025 Nadimpalli, Pattanayak and Prasad [28] identified the nonzero values themselves, for elements of order dividing the Coxeter number, as *dimensions* of representations of another group. Fifty years, one question, three sharper answers: *what is the value?*

Now step off that element. Once q no longer divides the Coxeter number no such description is available, the values are algebraic integers of no particular shape, and asking for them one at a time stops being the useful question. Ask instead what they *generate*. Fix g_q ; the values of all characters at g_q span a number field and generate inside it a subring A of finite index in the ring of integers $\mathcal{O}$ — that is, an *order*, and the invariant that measures an order is not new either. It is Dedekind’s conductor, introduced in 1877 precisely to say how far an order stands from the maximal one and how their ideal classes differ [4]. So the question *which ring do the*

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character values generate? is a nineteenth-century question about orders, asked of generators supplied by representation theory — and the two halves have rarely been in the same room.

The question itself has seldom been put. For finite groups Bächle and Sambale [1] proved that every prime dividing the index divides the group order. For a compact Lie group at a single torsion element the evaluation map itself is classical: Pianzola studied the homomorphism from the representation ring to $\mathbb{Z}[\zeta_N]$ that it defines, and determined the *field* it spans through a decomposition group in the Weyl group [29, Prop. 10, Prop. 13(b)] — but not the ring. We have found the ring put only in [14], which computed the conductor for $\mathrm{Sp}(2m)$ in its main families — and found the answer to be arithmetic rather than Lie-theoretic. The index is read off a moment matrix built from Stickelberger’s sawtooth; its maximal minors are Sinnott’s index of the Stickelberger ideal [30], and the relative class number $h^-(\mathbb{Q}(\zeta_q))$ governs whether the first layer is full — together, when the segment covers whole periods, with the Euler factor $\chi(2) - 1$ [14, Cor. 17], which is the subject of [32]. That arithmetic is why class numbers run through §§3–6.

There is a third question, and it was answered long ago in a room neither of the other two was in. *When do two different groups take the same values?* Frenkel asked it for affine Lie algebras in 1982 [24]; Pressley and Segal have it as a symmetry $k \leftrightarrow N - k$ of loop group representations; Nakanishi and Tsuchiya proved it for WZW models; Witten derived it from the one line of linear algebra that makes it obvious, $G_V(k, N) \cong G_{V^*}(N - k, N)$ [26, 20, 27]. It is level-rank duality, and in forty years it has been stated for fusion algebras, quantum dimensions, conformal blocks, link invariants and vector bundles on curves — and never, as far as we have found, for the ring the values generate.

That is the shape of this note. The three questions are about the same object, and the answers sit one on top of another: level-rank says *which* segments give the same values, the value lane says *what* those values are, and the conductor says *what they generate*. The division lemma of §2 is one line long and it crosses all three: it reproves the pairing, it adds two more operations that are not pairings, and it delivers the answer in the currency of the third lane, orders and conductors. Figure 1 is that account drawn: three lanes, fifty years, and the single claim this note makes about the picture as a whole.

So: what happens in the other classical types?

The first answer is a coincidence. The index for $\mathrm{SU}(n)$ gives, cell after cell, *the same integers* as the published table for $\mathrm{Sp}(2m)$ — not similar numbers, the same ones. The explanation is one line long, and it is Proposition 2(a) below: the two rings are equal, because the two eigenvalue segments differ by a single 1, and that is a monic integer factor.

That coincidence is the point of this note. If two different groups give the same ring, then the ring is not an invariant of the group. It is an invariant of the *segment* of eigenvalue exponents — and not even of that, but of the segment up to three explicit operations. One of the three is level-rank duality, which has been known for forty years; the point of Lemma 1 is not that it proves that one, but that it proves all three at once and in a form — an equality of orders — that the fusion-algebra literature has no reason to state. The classical groups on the ρ -line realise the same block of residues in only three positions.

If the position of the block is what matters, which positions can produce a defect at all, how large is it, and what decides its size?

The table answers them, one row per section, and adds the answer the question did not ask for: the uncentred position is not a regime of its own either, but sits over the first as a free extension of degree 2. As in [14], a statement is either proved or marked *measured* with the number of cells behind it; what is left open is marked $\triangleright$ where it arises, and the two that outlive their own section are collected at the end.

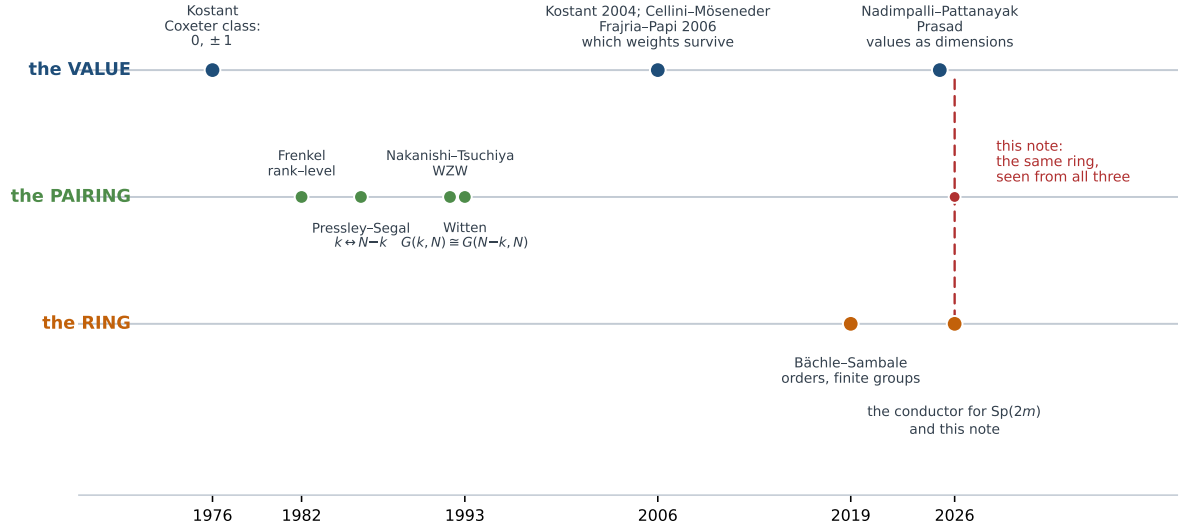


Figure 1: Three lanes, fifty years, and this note at the crossing. *What is the value?* was answered three times, each time better; the middle milestone is two papers, Kostant’s criterion of 2004 and its affine-Weyl description in the classical types of 2006. *When do two groups give the same values?* was answered between 1982 and 1993 and is level-rank duality; it is Proposition 2(b) below, in other words. *What ring do the values generate?* was put once for finite groups and once for a compact group. The dashed line is the only claim this note makes about the picture as a whole: that the three lanes are three views of one ring.

2 The segment, and the three operations that preserve the ring

Write the eigenvalues of g_q as $x_j = \zeta_M^{a_j}$. The multiset $\{a_j\}$ is the *segment*. For $\mathrm{Sp}(2m)$ at the principal element of type ρ it is $\{\pm 1, \dots, \pm m\}$ with $M = q$. For $\mathrm{SU}(n)$, with $\rho = (\frac{n-1}{2}, \dots, -\frac{n-1}{2})$, the entries are integers when n is odd and half-integers when n is even, so

$$\{-m, \dots, m\} \quad (n = 2m + 1, \quad M = q), \quad \{\pm 1, \pm 3, \dots, \pm(n-1)\} \quad (n \text{ even}, \quad M = 2q).$$

In both cases the segment is symmetric, so g_q is conjugate to its inverse and every character value is real; A is an order in the real subfield. It is convenient to write both cases on the same circle: with $t = \zeta_{2q}$ the segment of $\mathrm{SU}(n)$ is

$$\{n-1, n-3, \dots, 1-n\} \subset \mathbb{Z}/2q,$$

n terms of one parity class modulo $2q$ — so for n even g_q has order $2q$ in $\mathrm{SU}(n)$, and order q only in the adjoint group; we keep q as the parameter — and the characteristic polynomial of g_q in the standard representation is $F_{n,q}(T) = \prod_{j=0}^{n-1} (T - t^{n-1-2j})$.

Since the fundamental characters of $\mathrm{SU}(n)$ are, up to a unipotent triangular change over $\mathbb{Z}$, the elementary symmetric functions of the eigenvalues, and $e_n = \prod_j x_j = 1$, the ring A is generated by the coefficients of $F_{n,q}$. Everything in this section is then a statement about polynomials, and one lemma proves all of it.

Lemma 1 (division). *For a monic $F \in \mathcal{O}[T]$ let $\mathcal{A}(F) \subseteq \mathcal{O}$ be the subring generated by its coefficients. If F, G are monic and $FG \in \mathbb{Z}[T]$, then $\mathcal{A}(F) = \mathcal{A}(G)$. In particular, multiplying F by a monic factor of $\mathbb{Z}[T]$ does not change $\mathcal{A}(F)$.*

Proof. Dividing $FG \in \mathbb{Z}[T]$ by the monic F is a division inside $\mathcal{A}(F)[T]$: the quotient, which is G , has all its coefficients in $\mathcal{A}(F)$. Hence $\mathcal{A}(G) \subseteq \mathcal{A}(F)$, and exchanging F and G gives the

one block, three positions: the conductor depends on where the segment sits

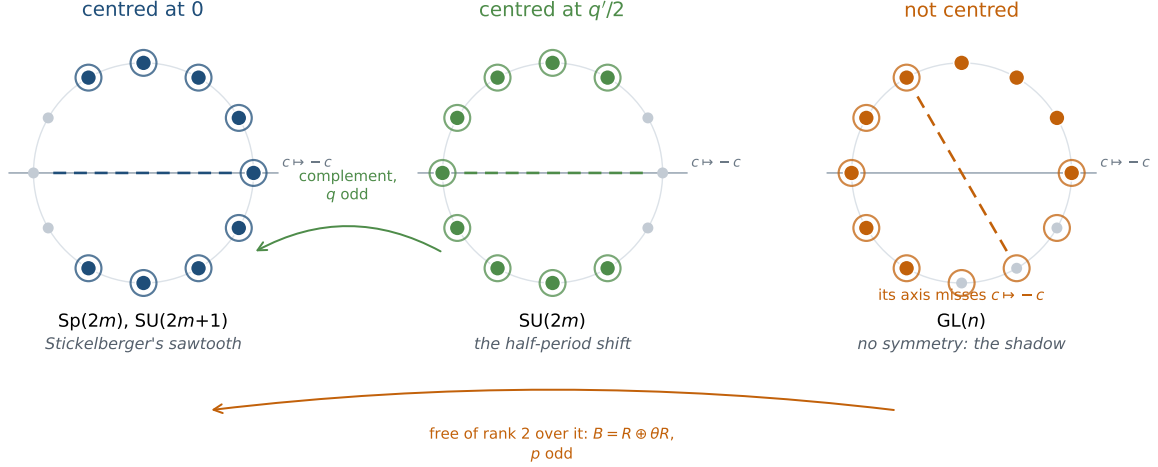


Figure 2: The same block of residues, in the three positions the classical groups realise. Filled dots are the classes the segment occupies; the hollow rings are their images under $c \mapsto -c$; the dashed line is the block's own axis, through its midpoint. When the block is centred at 0 or at $q'/2$ that axis is the grey axis $c \mapsto -c$, the rings fall on the dots, and the ring of character values is real; when it is not centred the two axes part, the rings come away from the dots, and the ring is not real. The green arrow is Corollary 3: for q odd the middle position is not a separate regime, because the complement carries it to the left one. The orange arrow is Theorem 25, and it is weaker in kind: the right position does not *become* the left one, but locally at an odd p and in the families $n = Nr, Nr + 1$ it is a free extension of rank 2 of it, $B = R \oplus \theta R$, where R is the ring of the segment recentred at $Nr/2$ — a segment of the leftmost kind. So the three positions are two, and the second sits over the first.

other inclusion. For the last sentence, if $H \in \mathbb{Z}[T]$ is monic then F and FH have the same coefficient ring: $\mathcal{A}(FH) \subseteq \mathcal{A}(F)$ is clear, and dividing FH by the monic integer polynomial H recovers F inside $\mathcal{A}(FH)[T]$. $\square$

The three operations are three ways of supplying such a factor.

Proposition 2 (the three operations). *Fix q and write $A_{\mathrm{SU}(n)}(q)$ for the ring generated by the character values of $\mathrm{SU}(n)$ at g_q . Then*

- (a) adjoining an eigenvalue 1: $A_{\mathrm{SU}(2m+1)}(q) = A_{\mathrm{Sp}(2m)}(q)$;
- (b) the complement: $A_{\mathrm{SU}(n)}(q) = A_{\mathrm{SU}(q-n)}(q)$ for $1 \leq n < q$;
- (c) the period: $A_{\mathrm{SU}(n+q)}(q) = A_{\mathrm{SU}(n)}(q)$.

Proof. (a) The segment of $\mathrm{SU}(2m + 1)$ is that of $\mathrm{Sp}(2m)$ together with one extra eigenvalue equal to 1, so the two characteristic polynomials differ by the monic factor $T - 1 \in \mathbb{Z}[T]$, and Lemma 1 applies. (The generators of $\mathrm{Sp}(2m)$ are its own fundamental characters, again a triangular change from the e_k of its $2m$ eigenvalues.)

(b) The q exponents $n - 1 - 2j$, $0 \leq j \leq q - 1$, run over one full parity class modulo $2q$, so

$$\prod_{j=0}^{q-1} (T - t^{n-1-2j}) = T^q - (-1)^{n-1} \in \mathbb{Z}[T].$$

question	answer
Why do two different groups give the same ring? (§2)	because they give the same segment up to a monic integer factor: three operations, one division lemma. The middle one is level-rank duality, known since [26, 20]; the other two are not, and the same lemma proves all three. Lemma 1, Figure 3
What does the defect look like at one prime? (§3)	a single residue field, a first layer measured by a cyclic module, and when $e = 2$, $v_p = 2 \dim - 1 - W_1$: Theorem 10, Figure 4
Which primes can occur at all? (§4)	only those with $r \mid n - 1, n, n + 1$: inside the families the defect is forced once $\varphi(q') \geq 4$ (Proposition 12), outside them the order is maximal, by a lemma about intervals of residues and a count of moments at the first layer: Lemma 13, Proposition 14, Theorem 16
And when the segment is not centred? (§6)	the same picture with the layer dimension doubled, generators being twisted Gaussian binomials — and, in the families $n = Nr, Nr + 1$ with $2 \leq n < q$, a free quadratic extension $B = R \oplus \theta R$ of the ring of the recentred segment, its layers being those of R added to themselves shifted by P and its type equal: Lemma 21, Theorem 22, Theorem 25, Proposition 28, Figure 5
Why do class numbers appear? (§§3, 6)	h^- governs whether the first layer is full — where the hypotheses of Theorem 10 and Proposition 23 hold, and only as a measurement in the even regime, q even and p odd — and in the shadow whether it reaches its reference rank $d + 1$, with the Euler factor at 2 as a second source when the segment covers whole periods: Theorem 10, Proposition 23, Theorem 25

The first n factors are $F_{n,q}$. In the remaining $q - n$ the exponents are $-(n+1) - 2i$, $0 \leq i \leq q - n - 1$, which is the segment of $\mathrm{SU}(q - n)$ translated by q ; translating by q multiplies every eigenvalue by $t^q = -1$, and $e_k(-x) = (-1)^k e_k(x)$, so that factor is $(-1)^{q-n} F_{q-n,q}(-T)$ and has the same coefficient ring as $F_{q-n,q}$. Now apply Lemma 1.

(c) The segment of $\mathrm{SU}(n + q)$ is one full parity class together with the segment of $\mathrm{SU}(n)$ translated by q ; the first contributes the monic integer factor $T^q - (-1)^{n+q-1}$ and the second, as in (b), has the coefficient ring of $\mathrm{SU}(n)$. $\square$

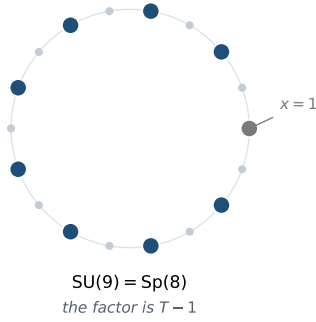
Part (b) is not new, and its combinatorial reason is the one everybody uses: choosing n of q things is choosing the other $q - n$. With $q = n + k$ it is the level-rank pairing $\mathrm{SU}(n)_k \leftrightarrow \mathrm{SU}(k)_n$. Witten derives it from $G_V(k, N) \cong G_{V^*}(N - k, N)$ and attributes the $k \leftrightarrow N - k$ symmetry of loop group representations to Pressley and Segal [27, §3], [26, Prop. (10.6.4)]; the explicit relation between the modular matrices of $\widehat{\mathfrak{su}}(N)_K$ and $\widehat{\mathfrak{su}}(K)_N$ is [19, (3.8)], whose row $R' = 0$ says that the quantum dimensions agree; Naculich and Schnitzer attribute it to earlier work [21, 22, 23], and the proof for WZW models is [20]. What (b) adds is the formulation as an equality of *orders* — that literature asks about fusion algebras and quantum dimensions, not about the ring the values generate — and a proof that is three lines of polynomial algebra, with no fusion category, no modular matrices and no loop group. The same three lines give (a) and (c), which are not level-rank.

Measured: Not the ring but the *values*: computing $\chi_\lambda(g_q)$ by the bialternant in $\mathbb{Q}(\zeta_{2q})$ over the whole alcove, the sets of character values of $\mathrm{SU}(n)$ and of $\mathrm{SU}(q - n)$ at g_q coincide exactly — not merely up to sign — in 25 of 25 pairs with $q \leq 12$, up to 462 values a side. That is sharper than the equality of rings, and it is what identifies (b) with level-rank rather than merely with something like it. For the normalised quantum dimensions the row $R' = 0$ of [19, (3.8)] already

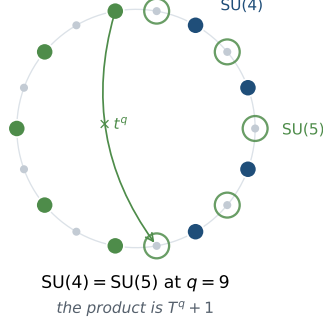
gives it; the measurement is a control of our conventions, not a substitute for that proof.

three operations on the segment, one division lemma: the ring does not change

(a) adjoin the eigenvalue 1



(b) take the complement: level-rank



(c) one full turn

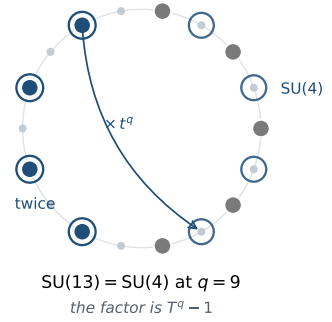


Figure 3: The three operations of Proposition 2 on the circle of $2q$ -th roots of unity, drawn at $q = 9$. The segment of $SU(n)$ is n consecutive exponents of one parity class. (a) Adjoining the eigenvalue 1 is the factor $T - 1$. (b) The n exponents and the remaining $q - n$ multiply to $T^q + 1$, and the leftovers are the segment of $SU(q - n)$ turned half a turn, which is multiplication by $t^q = -1$; this panel is level-rank duality, and the half turn is all that the division lemma has to absorb. (c) Thirteen exponents at $q = 9$ cover the class once and four of them twice; the full class is the factor $T^q - 1$ and the four that repeat are again $SU(4)$ up to t^q .

Measured: 287 cells with $m \leq 7$, $q \leq 49$ for (a), comparing the two *lattices* and not merely their indices; 90 complements and 82 periods, also as lattices, with $q \leq 21$ and $\varphi(2q) \leq 24$. No discrepancy.

Two consequences deserve to be stated separately, because they remove a case rather than add one.

Corollary 3. *Let q be odd and $n < q$ even. Then $A_{SU(n)}(q) = A_{Sp(q-n-1)}(q)$; for $n > q$ apply Proposition 2(c) first.*

Proof. $q - n$ is odd, so Proposition 2(b) and then (a) apply. □

So for q odd the even case is not a separate theory: it is the type- C ring at the same q , on the nose and not merely up to conjugacy. What survives of §5 is the case q even.

Remark 4 (the even segment is $Sp(2m)$ at the other principal element). The segment $\{\pm 1, \pm 3, \dots, \pm(2m - 1)\}$ of $SU(2m)$ is also the segment of $Sp(2m)$ at $\exp(2\pi i \rho^\vee / q)$, since $\rho^\vee = (m - \frac{1}{2}, \dots, \frac{1}{2})$ for C_m gives exactly the odd exponents modulo $2q$. As the spectrum is inverse-closed, the e_k of Sp and of $SU(2m)$ generate the same ring. The two principal elements ρ and $\rho^\vee$ of $Sp(2m)$ are therefore the two symmetric positions of Figure 2, and type A adds no third one.

Remark 5 (why Sp had to appear). The pair $(SU(2m + 1), Sp(2m))$ is the folding $A_{2m} \rightsquigarrow C_m$: the fixed group of the diagram automorphism of $SL(2m + 1)$ is $SO(2m + 1)$, whose dual is $Sp(2m)$; and g_q , having inverse-closed spectrum and determinant 1, lies up to conjugacy in that fixed group. Jantzen's twining character formula [7, 6, 5] lives in the same picture, but it equates a *twisted* trace with an ordinary character of the dual of the fixed group, whereas Proposition 2(a) equates ordinary values with ordinary values and is proved by a division. We cite it as context, not as precedent.

3 The local picture

Fix p and write $M = p^v q'$ with $p \nmid q'$, $E = \varphi(p^v)$, $d = \varphi(q')/2$. We assume $n \not\equiv 0, \pm 1 \pmod{q}$, since otherwise $A = \mathbb{Z}$ by Proposition 2 and there is nothing to compute, and $q' \geq 3$: for $q' \leq 2$ the residual algebra is $\mathbb{F}_p$, d must be read as 1, and the matrix of Theorem 10(b) has no columns — at $\mathrm{SU}(3)$, $q = 8$, $p = 2$ it has rank 0 while $W_1 = 1$. In $\mathbb{Z}_p[\zeta_M]$ write $\zeta_M = \omega \rho$ with ω of order q' and ρ of order p^v , and put $\pi = \rho - 1$, a uniformiser, $\gamma_c = \omega^c$, and

$$N_c = \#\{j : a_j \equiv c \pmod{q'}\}, \quad S(c) = \sum_{a_j \equiv c} a_j.$$

By the symmetry of the segment $S(-c) = -S(c)$; in particular $S(0) = 0$, and $S(q'/2) = 0$ when q' is even.

Lemma 6 (first order). *With $F(T) = \prod_c (T - \gamma_c)^{N_c}$,*

$$\prod_j (T - x_j) = F(T) - \pi \sum_c S(c) \gamma_c \frac{F(T)}{T - \gamma_c} + O(\pi^2).$$

Proof. Since $\rho = 1 + \pi$, each eigenvalue is $x_j = \omega^{a_j} \rho^{a_j} = \gamma_{a_j} (1 + a_j \pi + O(\pi^2))$, so $T - x_j = (T - \gamma_{a_j}) - a_j \gamma_{a_j} \pi + O(\pi^2)$. Expanding the product over j and keeping the term linear in π ,

$$\prod_j (T - x_j) = \prod_j (T - \gamma_{a_j}) - \pi \sum_j a_j \gamma_{a_j} \prod_{i \neq j} (T - \gamma_{a_i}) + O(\pi^2),$$

and $\prod_{i \neq j} (T - \gamma_{a_i}) = F(T)/(T - \gamma_{a_j})$, an exact division in $\mathcal{O}[T]$. Grouping the indices j by the class $c = a_j \pmod{q'}$ turns $\sum_j a_j \gamma_{a_j} F(T)/(T - \gamma_{a_j})$ into $\sum_c S(c) \gamma_c F(T)/(T - \gamma_c)$. $\square$

The lemma says that the first-order parts of *all* the generators at once are the coefficients of a single polynomial. They are not independent data: in the stable case (Definition 8) — residue scalar and $E \geq 2$, the setting of Theorems 10 and 22 — $\mathrm{gr}_1(A)$ is their $\mathbb{F}_p$ -span; without stability the residual algebra multiplies them, and at $\mathrm{GL}(2)$, $q = 15$, $p = 3$ they span one direction over $\mathbb{F}_3$ while the layer is full, of dimension 4. It is stated in the eigenvalues x_j , which is the right model for the non-real ring of §6; for a symmetric segment one has to pass to real coordinates first, and that is where the quantum integers of [14] come from.

Lemma 7 (real coordinates). *Let the segment be symmetric and write $\eta_j = x_j + x_j^{-1}$, $z = (\omega - \omega^{-1})\pi$ and*

$$U_c = \frac{\omega^c - \omega^{-c}}{\omega - \omega^{-1}},$$

the quantum integer, which satisfies $\sigma_u(U_c) = U_{uc}/U_u$ and vanishes exactly when $2c \equiv 0$. Write also $\bar{\gamma}_c = \gamma_c + \gamma_c^{-1}$, the real counterpart of γ_c , which still satisfies $\sigma_u(\bar{\gamma}_c) = \bar{\gamma}_{uc}$. Then $A = \mathbb{Z}[e_k(\eta)]$, each $\eta_j \equiv \bar{\gamma}_{a_j} + a_j U_{a_j} z$ modulo $\mathfrak{r}^2$, and

$$\prod_j (T - \eta_j) \equiv F_\eta(T) - z G(T), \quad G(T) = \sum_c S(c) U_c \frac{F_\eta(T)}{T - \bar{\gamma}_c},$$

where c runs over one representative of each pair $\{c, -c\}$ of classes met by the half segment, both in the sum and in $F_\eta = \prod_c (T - \bar{\gamma}_c)^{N_c}$, and N_c counts the $a_j > 0$ with $a_j \equiv \pm c$. Summing $S(c) U_c$ over both members of a pair would count each term twice, since $S(-c) = -S(c)$ and $U_{-c} = -U_c$.

Proof. For a symmetric segment $\prod_j (T - x_j) = (T - 1)^\varepsilon \prod_{j>0} (T^2 - \eta_j T + 1)$ with $\varepsilon \in \{0, 1\}$. The factor $(T - 1)^\varepsilon$ is monic in $\mathbb{Z}[T]$, so by Lemma 1 it may be dropped; and the coefficients of the remaining product are $\mathbb{Z}$ -polynomials in the $e_k(\eta)$, triangularly, and conversely, so the ring is $\mathbb{Z}[e_k(\eta)]$. The expansion is Lemma 6 applied to the η_j : from $x_j = \gamma_{a_j}(1 + a_j\pi + O(\pi^2))$,

$$\eta_j = \gamma_{a_j} + \gamma_{a_j}^{-1} + a_j\pi(\gamma_{a_j} - \gamma_{a_j}^{-1}) + O(\pi^2) = \bar{\gamma}_{a_j} + a_j U_{a_j} z + O(\pi^2),$$

and grouping the $j > 0$ by pairs $\{c, -c\}$, whose terms $a_j U_{a_j}$ add up to $S(c)U_c$, gives G . Note that z , not π , is the uniformiser of the real ring — up to a term of order π^2 : z itself is not real, its conjugate being $z/(1 + \pi)$, and the real element $z_0 = \zeta_M + \zeta_M^{-1} - (\omega + \omega^{-1})$ agrees with it modulo π^2 , which is all that is used. The two differ by the unit $\omega - \omega^{-1}$, and it is exactly that unit which turns the non-real coefficient γ_c of Lemma 6 into the real U_c . Writing G with γ_c in place of U_c , or with $T - \gamma_c$ in place of $T - \bar{\gamma}_c$, would be a different polynomial: the two models must not be mixed. $\square$

Definition 8. The segment is *stable* at q' if the multiset $\{a_j \bmod q'\}$ is invariant under multiplication by every unit of $\mathbb{Z}/q'$.

Lemma 9. *The segment is stable at q' if and only if N_c depends only on $\gcd(c, q')$.*

Proof. The orbits of $(\mathbb{Z}/q')^\times$ on $\mathbb{Z}/q'$ are the sets $\{c : \gcd(c, q') = D\}$, one for each divisor D of q' . A multiset is invariant exactly when its multiplicity function is constant on orbits. $\square$

We write $\mathfrak{r}$ for the radical of $\mathcal{O}_p = \mathcal{O} \otimes \mathbb{Z}_p$ (for the real ring, of the real $\mathcal{O}_p$), $\bar{R} = \mathcal{O}_p/\mathfrak{r}$ for the residual algebra, of dimension d over $\mathbb{F}_p$, and $W_i = \dim_{\mathbb{F}_p}((A_p \cap \mathfrak{r}^i) + \mathfrak{r}^{i+1})/\mathfrak{r}^{i+1}$ for the dimension of the i -th layer of A ; e is the least i with $A_p \supseteq \mathfrak{r}^i$, the exponent of the conductor.

Theorem 10 (local structure; [14, Thm. 5]). *Assume the segment is stable at q' and $E \geq 2$. Then*

- (a) $F_\eta \in \mathbb{F}_p[T]$, the image of A in $\bar{R}$ is $\mathbb{F}_p$ — that is, $W_0 = 1$ — and A has a single maximal ideal above p , with residue field $\mathbb{F}_p$;
- (b) $W_1 = \dim_{\mathbb{F}_p} \mathbb{F}_p[(\mathbb{Z}/q')^\times] \cdot S$, the rank modulo p of $[S(u^{-1}c)]$ with rows $u \in (\mathbb{Z}/q')^\times / \pm 1$ and columns the classes with $2c \not\equiv 0$;
- (c) if moreover $d \geq 2$, then $e = 1$ if and only if $W_1 = d$;
- (d) $v_p([\mathcal{O} : A]) = \sum_{i < e} (d - W_i)$; in particular $v_p = d - 1$ when $e = 1$ and $v_p = 2d - 1 - W_1$ when $e = 2$.

Proof. (a) Stability makes the Galois group permute the factors of F_η , so its coefficients are fixed and $F_\eta \in \mathbb{F}_p[T]$; the e_k reduce to those constants in every residue field.

(b) Every element of A is $\Phi(e)$ with $\Phi \in \mathbb{Z}[X]$, and modulo $\mathfrak{r}^2$,

$$\Phi(e) \equiv \Phi(F_\eta) + z \sum_k (\partial_k \Phi)(F_\eta) G_k,$$

with $\Phi(F_\eta) \in \mathbb{F}_p$ and $p \in \mathfrak{r}^2$ because $E \geq 2$. So $\text{gr}_1(A)$ is the $\mathbb{F}_p$ -span V of the coefficients G_k , which by Lemma 7 are those of G . As $\bar{R}$ is étale, $\bar{R} \otimes \bar{\mathbb{F}}_p \cong \bar{\mathbb{F}}_p^d$ through the embeddings σ_u , and extension of scalars is injective on $\mathbb{F}_p$ -subspaces, so $\dim V$ is the rank of the matrix whose row u lists the coefficients of $\sigma_u(G)$. From $\sigma_u(\bar{\gamma}_c) = \bar{\gamma}_{uc}$, $\sigma_u(U_c) = U_{uc}/U_u$ and the stability of F_η ,

$$U_u \sigma_u(G) = \sum_c S(u^{-1}c) U_c \frac{F_\eta(T)}{T - \bar{\gamma}_c},$$

and multiplying a row by the unit U_u does not change the rank; the functions $U_c F_\eta / (T - \bar{\gamma}_c)$, one for each class c up to sign, are independent and do not depend on u — the classes c and $-c$ give opposite functions and columns that agree up to sign, so keeping both does not change the rank — and the classes with $2c \equiv 0$ contribute nothing because there $U_c = 0$ (and $S(c) = 0$).

(c) If $\text{gr}_1(A) = \text{gr}_1(\mathcal{O})$ then $\text{gr}_i(A) \supseteq \text{gr}_1(A)^i = \text{gr}_i(\mathcal{O})$ for all i , since $\text{gr}(\mathcal{O}_p)$ is a polynomial ring over $\bar{R}$, and the filtration collapses, so $e \leq 1$; and $e \geq 1$ because $W_0 = 1 < d$ makes $A_p \neq \mathcal{O}_p$. The converse is clear.

(d) Filter $\mathcal{O}_p$ and A_p by $\mathfrak{r}^i$; for $i \geq e$ one has $A_p \cap \mathfrak{r}^i = \mathfrak{r}^i$, so only the layers $i < e$ contribute, with quotients of dimension $d - W_i$. The layer $i = 0$ always contributes $d - 1$, by (a). $\square$

Measured: The instrument reproduces the published type- C law in 18 of 18 cells. In type A , (d) with $e = 2$ holds in 44 of 44 cells under $W_1 > d/2$; of the 18 cells outside the hypothesis, 15 still agree and the 3 that fail all have $W_1 = 0$, the ladder zone of [14]. Statement (b) holds in 193 of 193 stable cells and in 98 of 127 unstable ones: the hypothesis is not decoration. That $W_0 = 1$ is not assumed but read off the instrument in every cell tabulated in §4.

The last clause of (d) is worth saying out loud, because it is what §4 rests on: under the hypotheses of the theorem the defect is never zero once $d \geq 2$. A full first layer makes $e = 1$, not $v_p = 0$.

That $d \geq 2$ is not decoration either, and it is why (c) carries it while the other three parts do not. At $\text{SU}(3)$ with $q = 12$ and $p = 3$ the segment $\{1, 0, -1\}$ is stable at $q' = 4$ with $E = 2$, and $d = 1$, so $W_1 = 1 = d$ — and yet $A = \mathbb{Z}[\sqrt{3}] = \mathcal{O}$ is maximal, $\mathfrak{f} = \mathcal{O}$ and $e = 0$. What fails is not the layer count but the step before it: $W_0 = 1$ stops forcing a defect once $d = 1$, and then $e = 1$ has nothing to be. Parts (a), (b) and (d) stand as written, (d) reading as the empty sum. The source keeps the same corner out by excluding $q' \in \{1, 2, 3, 4, 6\}$ among its standing hypotheses [14, §5].

Statement (d) is a plane, and it is worth seeing it as one. Over a wider sweep of 417 cells the real ring lives with $d = \varphi(q')/2$ and the shadow of §6 with $D = \varphi(q')$; if the layer dimension is put on an axis, *both* populations lie on the same plane $v = 2 \dim - 1 - W_1$. That is the thesis of this note in one picture: the group does not enter, only the segment does.

The plane is the face $e = 2$ of (d), and it is worth knowing what lies off it. In the regime of Theorem 25 with $p \nmid N$ and the real profile of [14] (the display after Proposition 26), writing $\delta = d - W_1(R)$ for the drop of the real first layer, the shadow has $W_1 = d + 1 - \delta$ and $\dim = 2d$, so the plane would predict $3d - 2 + \delta$ while the true index is $3d - 2 + 2\delta$: a cell of the shadow sits *exactly* δ above the plane, and the height off the plane is the class-number defect. With $p \mid N$ the count is different: at $q = 45$, $p = 3$, $N = 3$ the shadow has $W_1 = 0$ and $v_3([\mathcal{O} : B]) = 14$, where this reading would give 1 and 8. No cell of the sweep shows this — its 14 off-plane shadow cells all have $p = 2$ or $(p, q') = (3, 4)$, outside that regime, and the sweep never reaches the first conductor with $\delta > 0$ — so the figure is a picture of the face $\delta = 0$.

4 Support: which primes can occur

Throughout this section write

$$r = \frac{q}{p^{v_p(q)}},$$

the prime-to- p part of q — not of M . For n odd, and for n even with $p = 2$, one has $r = q'$. For n even and p odd, $M = 2q$ gives $q' = 2r$: the segment then occupies the odd residues modulo $2r$, which as an arithmetic progression has period r , and r is the modulus that the criterion sees.

Proposition 11. *If r divides $n - 1$, n or $n + 1$, then the segment of $\text{SU}(n)$ is stable at q' .*

Proof. By Lemma 9 it suffices to see that N is constant on the orbits of $(\mathbb{Z}/q')^\times$. There are three branches. For n odd, $M = q$ and $q' = r$, and the segment $\{-m, \dots, m\}$ is n consecutive

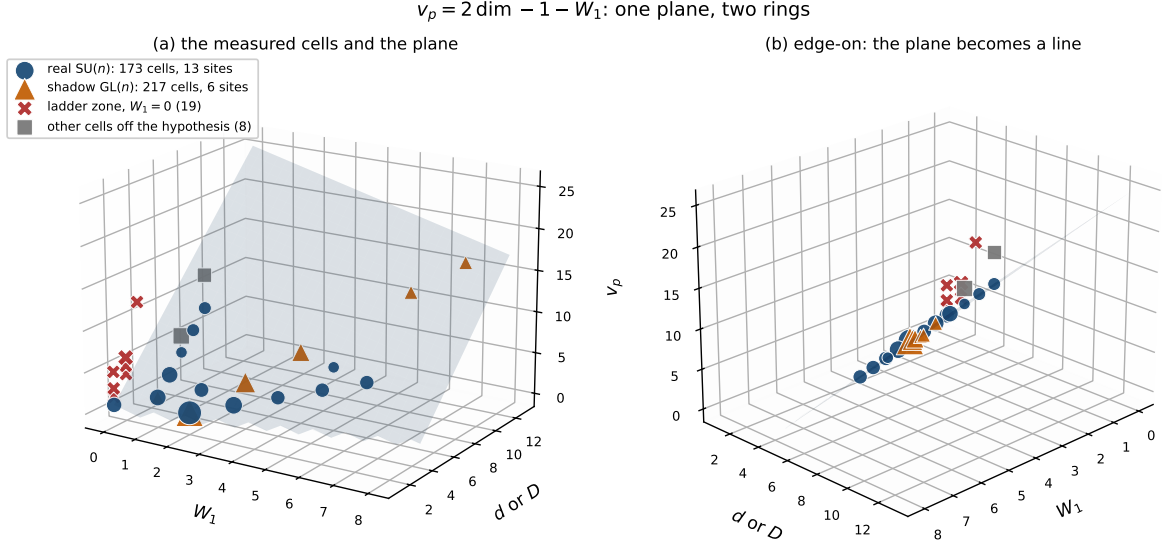


Figure 4: Each measured cell is a point $(W_1, \dim, v_p)$; many cells share a position, and the size of a marker says how many. *Left*: the cells and the plane. *Right*: seen along $(1, 1, 1)$, which is orthogonal to the normal $(1, -2, 1)$, the plane projects to a line and whatever lies on it becomes collinear. Under the hypothesis $W_1 > \dim/2$ all 335 cells lie on the plane; the 27 that do not are shown apart, 19 of them in the ladder zone $W_1 = 0$. All 14 off-plane cells of the shadow have $p = 2$ or $(p, q') = (3, 4)$; none is in the regime of Theorem 25, where the shadow leaves the plane upwards by exactly δ .

residues of $\mathbb{Z}/r$. For n even it is the progression $\pm 1, \pm 3, \dots, \pm(n-1)$ of difference 2 modulo $M = 2q$: with $p = 2$ one has $q' = r$ odd and the difference is invertible, and with p odd one has $q' = 2r$ and the progression runs through the odd classes, returning to the start after r steps. In every branch the progression has period r , so the classes it does not meet are either none or all the even ones, and those are permuted among themselves by the units, so they contribute nothing. On the classes it meets, $N_c \in \{\lfloor n/r \rfloor, \lceil n/r \rceil\}$, and the two values coincide when $r \mid n$. If $n \equiv \pm 1 \pmod{r}$, exactly one class c is off by one, and since the segment is symmetric $N_{-c} = N_c$, so that class satisfies $c \equiv -c$. Such a class is fixed by every unit, because the orbit of c has $\varphi(q'/\gcd(c, q'))$ elements and that is 1 exactly when $2c \equiv 0$. So N is constant on orbits. (For n odd the exceptional class is not always the centre 0: writing $n = Ar \pm 1$ it is $Ar/2 \pmod{r}$, which is $q'/2$ when A is odd — at $SU(5)$ with $q' = 4$ it is 2.) $\square$

Measured: 807 cells with $n \leq 25$, $q \leq 60$, over the four branches (both parities of n and of r): every cell in a family is stable, with no exception.

The converse fails: there are stable segments outside the families, the smallest being $n = 3$, $q' = 6$, whose residues $\{0, 1, 5\}$ are the union of the orbits $\{0\}$ and $\{1, 5\}$. Every one of them has $q' = 6$, as Lemma 13 will force, and they are maximal for the reason given in the proof of Proposition 14: there $d = 1$ and $W_1 = 1$. It is not $d = 1$ alone: inside the families, $d = 1$ can carry a defect.

Inside the families the defect is not merely possible: it is forced, and by the local law itself.

Proposition 12 (the families always give a defect). *Let $E \geq 2$, let the segment be stable at q' — by Proposition 11 it is enough that r divide one of $n-1, n, n+1$ — and let $\varphi(q') \geq 4$, that is $q' \notin \{1, 2, 3, 4, 6\}$, that is $d \geq 2$. Then*

$$p \mid [\mathcal{O} : A], \quad \text{indeed } v_p([\mathcal{O} : A]) \geq d - 1.$$

Proof. Theorem 10 applies. By (a) the layer $i = 0$ contributes $d - W_0 = d - 1$ to the sum in (d), and every term of that sum is ≥ 0 . $\square$

So inside the families a defect can be avoided only in the range $\varphi(q') \leq 2$, where $d = 1$ and the argument says nothing: the layer $i = 0$ is then full by default, and a defect, if there is one, sits in the deeper layers. It can: $\text{SU}(11)$ at $q = 36$ and $p = 3$ is in the family $r = 4 \mid 12$, has $d = 1$, and has index 3, with profile $(1, 0, 1, \dots)$.

Measured: 26 of 26 stable cells with $E \geq 2$ and $d \geq 2$, over $n \leq 7$ and $q \leq 40$, have $v_p \geq d - 1$; restricted to the cells in a family with $r \notin \{1, 2, 3, 4, 6\}$, 23 of 23 have $p \mid [\mathcal{O} : A]$.

Lemma 13 (stable intervals). *Let $q' \geq 3$ and let $I \subseteq \mathbb{Z}/q'$ be a set of B consecutive residues with $2 \leq B \leq q' - 2$. If $\alpha(I) = I$ for one affine bijection $\alpha(x) = ux + b$ of $\mathbb{Z}/q'$, then $u \equiv \pm 1 \pmod{q'}$. In particular, if I is a union of the strata $X_D = \{c : \gcd(c, q') = D\}$ then $\varphi(q') \leq 2$; indeed the only such I occur at $q' = 6$, $B = 3$, and they are $\{5, 0, 1\} = X_1 \cup X_6$ and $\{2, 3, 4\} = X_2 \cup X_3$.*

Proof. The complement of a proper cyclic interval is a cyclic interval, and it is α -stable when I is; so we may replace I by the shorter of the two, an interval J of length b with $2 \leq b \leq q'/2$ and $\alpha(J) = J$. Put $C_J(t) = |J \cap (J + t)|$. The bijection α carries $J \cap (J + t)$ onto $\alpha(J) \cap (\alpha(J) + ut) = J \cap (J + ut)$, so $C_J(ut) = C_J(t)$, and in particular $C_J(u) = C_J(1) = b - 1$. On the other hand C_J does not change when J is translated, and for $J = \{0, \dots, b - 1\}$ and $1 \leq t < q'$

$$C_J(t) = \max(0, b - t) + \max(0, b - (q' - t)).$$

Since $2b \leq q'$ the two terms are never positive together, and $b - 1 \geq 1$; so $C_J(t) = b - 1$ exactly when $t = 1$ or $t = q' - 1$, and $u \equiv \pm 1$.

If I is a union of strata it is stable under every unit, because multiplying by a unit does not change $\gcd(c, q')$. So every unit is ± 1 , that is $\varphi(q') \leq 2$ and $q' \in \{3, 4, 6\}$. At $q' = 3$ the range of B is empty. Writing $I = \{a, \dots, a + B - 1\}$, stability under -1 reads $2a + B - 1 \equiv 0 \pmod{q'}$, which for q' even forces B odd: that excludes $q' = 4$, where $B = 2$, and leaves $q' = 6$, $B = 3$, $a \in \{2, 5\}$. $\square$

Measured: The strong form by brute force: for every $q' \leq 120$, every interval with $2 \leq B \leq q' - 2$ and every unit u , $uI = I$ only for $u = \pm 1$ (3.1×10^7 checks), and the affine form for $q' \leq 60$ and every translation b (1.9×10^6 checks); and over all $q' \leq 150$, all B and all rotations, the only interval that is a union of strata outside $B \in \{0, 1, q' - 1\}$ is one of the two at $q' = 6$. The bound $B \geq 2$ is not decoration: the singleton $\{0\}$ is stable under every unit. That is the same $q' = 6$ as in the cells recorded below.

Proposition 14 (support). *Suppose r divides none of $n - 1$, n , $n + 1$. Then either the segment is stable at q' , and then $q' = r = 6$ and A is maximal at p ; or it is not, and then the image of A in $\bar{R}$ is all of $\bar{R}$, that is $W_0 = d$, and A is maximal at p when $v = 0$.*

Proof. Suppose r divides none of the three. As in the proof of Proposition 11, write the segment in order as $a_j = a_0 + \beta j$, $0 \leq j < n$, with $\beta = 1$ for n odd and $\beta = 2$ for n even; the class of a_j modulo q' depends on j only modulo r . Writing $n = Ar + B$, the multiplicity is $A + 1$ on the classes of the positions $J = \{0, \dots, B - 1\} \subseteq \mathbb{Z}/r$ and A on the others, and $B \notin \{0, 1, r - 1\}$, so $2 \leq B \leq r - 2$. A unit u of $\mathbb{Z}/q'$ sends a_j to $a_0 + \beta(uj + (u - 1)a_0/\beta)$ — for $\beta = 2$ the number a_0 is odd and u is odd — so on positions it acts as the affine bijection $\alpha_u(j) = uj + (u - 1)a_0/\beta$ of $\mathbb{Z}/r$. The map $u \mapsto \alpha_u$ is injective: u is determined by $u \bmod r$ unless $q' = 2r$ with r even, and then u and $u + r$ differ by the translation $r/2$.

If $v = 0$ the prime is unramified and $\mathcal{O}_p/p\mathcal{O}_p = \bar{R}$, so both cases below, which end with a full residual image, give $A_p = \mathcal{O}_p$ by Nakayama with no frontier. Suppose the segment stable at q' . Then $\alpha_u(J) = J$ for every unit, and Lemma 13 gives $u \equiv \pm 1 \pmod{r}$. An affine map with linear part 1 fixes the proper interval J only if it is the identity, since $C_J(t) < C_J(0)$ for $t \neq 0$,

and one with linear part -1 is then determined by J ; so the units give at most two maps, and $\varphi(q') \leq 2$. With $r \geq 4$, forced by $2 \leq B \leq r - 2$, and $q' \in \{r, 2r\}$ this leaves $q' = r \in \{4, 6\}$; $q' = r$ even means n odd, and at $r = 4$ that forces B odd, outside $[2, 2]$. So $q' = r = 6$, so n is odd, $B = 3$ and $n = 6A + 3$; and $p \geq 5$, so $E \geq 4$ when $v \geq 1$. There $d = 1$ and the residual image is all of $\bar{R} = \mathbb{F}_p$, which settles $v = 0$. For $v \geq 1$ the residue ring $\bar{R}$ is $\mathbb{F}_p$, and $\mathcal{O}_p$ is a discrete valuation ring totally ramified over $\mathbb{Z}_p$: it is generated over $\mathbb{Z}_p$ by any element of valuation one, so $A_p = \mathcal{O}_p$ as soon as $W_1 = 1$. By Theorem 10(b), W_1 is the rank of the single row $(S(1), S(2), -S(2), -S(1))$ over the classes 1, 2, 4, 5, and summing over $\{-m, \dots, m\}$ gives $(S(1), S(2)) = (A + 1, -A)$ for A even and $(A, -(A + 1))$ for A odd. Two consecutive integers are not both divisible by p ; so $W_1 = 1$ and there is no defect.

Otherwise the segment is unstable at q' , and more is true: by the same two steps a unit that fixes the multiset gives an α_u fixing J , hence $u \equiv \pm 1$ and α_u is one of at most two maps, so the stabiliser of the multiset in $(\mathbb{Z}/q')^\times / \pm 1$ is trivial. Then the image of A in $\bar{R}$ is all of $\bar{R}$, that is $W_0 = d$: that image is generated by the coefficients of F_η , and after extending scalars $\bar{R} \otimes \bar{\mathbb{F}}_p \cong \bar{\mathbb{F}}_p^d$ through the embeddings σ_u , a coefficient becomes the function $u \mapsto$ (that coefficient of $\sigma_u(F_\eta)$). As the $\bar{\gamma}_c$ are distinct for classes $c \not\equiv \pm c'$, two embeddings give the same polynomial only if $u'u^{-1}$ fixes the multiset, so these functions separate the d points, and functions separating the points of a finite set generate all functions on it. $\square$

A full residual image does not by itself make the order maximal: the index law of Theorem 10 reads $v_p([\mathcal{O} : A]) = \sum_{i < e} (d - W_i)$, and $W_0 = d$ kills the term $i = 0$ and no other. What closes the gap is that, once the residue field is reached, the first layer decides everything, and the first layer is read off the moments.

Proposition 15 (the first layer decides). *Let $E \geq 2$ and suppose the image of A in $\bar{R}$ is all of $\bar{R}$. Then $A_p = \mathcal{O}_p$ if and only if $p \nmid S(c)$ for some class c , taken with $2c \not\equiv 0$ when the segment is symmetric.*

Proof. The graded ring $\text{gr}(A)$ is a graded subring of $\text{gr}(\mathcal{O}_p) = \bar{R}[\varpi]$, polynomial in the class of a uniformiser, and $\text{gr}_0(A) = \bar{R}$ by hypothesis. So $\text{gr}_1(A)$ is an $\bar{R}$ -submodule of $\text{gr}_1(\mathcal{O}_p) = \bar{R}\varpi$ and $\text{gr}_i(A) \supseteq \text{gr}_1(A)^i$: if $\text{gr}_1(A) = \text{gr}_1(\mathcal{O}_p)$ every layer is full and $A_p = \mathcal{O}_p$, and conversely a maximal order has $W_1 = d$. It remains to identify $\text{gr}_1(A)$. Each generator is $e_k = e_k^{(0)} + g_k\varpi + O(\varpi^2)$, where $e_k^{(0)}$ is the corresponding coefficient of F (or F_η), which lies in the unramified subring $\mathbb{Z}_p[\omega]$, and g_k is the corresponding coefficient of G in Lemma 6 (or Lemma 7). An element $\Phi(e)$, $\Phi \in \mathbb{Z}_p[X]$, lies in $\mathfrak{r}$ exactly when $\Phi(\bar{e}) = 0$ in $\bar{R}$; then $\Phi(e^{(0)}) \in \mathfrak{r} \cap \mathbb{Z}_p[\omega] = p\mathbb{Z}_p[\omega] \subseteq \mathfrak{r}^E \subseteq \mathfrak{r}^2$, so the image of $\Phi(e)$ in gr_1 is $\sum_k (\partial_k \Phi)(\bar{e}) g_k$. As $\bar{R}$ is étale over $\mathbb{F}_p$, the vectors $((\partial_k \Phi)(\bar{e}))_k$ with $\Phi(\bar{e}) = 0$ span $\bar{R}^n$ — for $\bar{R} = \mathbb{F}_p[X]/I$ the conormal sequence gives $I/I^2 \cong \bar{R}^n$, because $\Omega_{\bar{R}/\mathbb{F}_p} = 0$. So $\text{gr}_1(A)$ is the $\bar{R}$ -span of the coefficients of G , and it is all of $\text{gr}_1(\mathcal{O}_p)$ exactly when G is nonzero in every factor of $\bar{R}$. In each factor, after cancelling F (or F_η) in the field of rational functions, the fractions $1/(T - \gamma_c)$ of distinct classes are independent, and γ_c and U_c ($2c \not\equiv 0$) are units; so G is nonzero in a factor exactly when some $S(c)$ is prime to p , a condition that does not depend on the factor. $\square$

Theorem 16 (support). *Let $2 \leq n \leq q - 2$. If p divides $[\mathcal{O} : A]$, then r divides $n - 1$, n or $n + 1$. This holds for every prime p , including $p = 2$.*

Proof. Suppose r divides none of the three. By Proposition 14, either $q' = r = 6$ and A is maximal at p , or the residual image is full. In the second case, if $E = 1$ the extension is unramified and Nakayama gives $A_p = \mathcal{O}_p$; if $E \geq 2$, Proposition 15 asks for one class with $p \nmid S(c)$. Keep the notation of the proof of Proposition 14: $a_j = a_0 + \beta j$, $n = Ar + B$ with $2 \leq B \leq r - 2$, the class of a_j depends only on $j \bmod r$ and does so bijectively onto the classes

the segment meets, with multiplicity $m_i = A + 1$ at the positions $i < B$ and A at the others. Summing each progression,

$$S_i = m_i(a_0 + \beta i) + \beta r \frac{m_i(m_i - 1)}{2}, \quad \text{so} \quad S_{i+1} - S_i = m\beta$$

whenever the positions i and $i + 1$ carry the same multiplicity m ; the pairs $(0, 1)$ and $(B, B + 1)$ are such pairs, one in each block. If p divided every $S(c)$ — including the classes with $2c \equiv 0$, where a symmetric segment has $S(c) = 0$ anyway — and $p \nmid \beta$, it would divide $(A + 1)\beta$ and $A\beta$, hence $A + 1$ and A . Now $\beta \in \{1, 2\}$, so $p \nmid \beta$ unless $p = 2$ and n is even; and then every exponent is odd, $S(c) \equiv N_c \pmod{2}$, and the two multiplicities A and $A + 1$ both occur, with opposite parities. Either way some $S(c)$ is prime to p . $\square$

So Proposition 12 and Theorem 16 together say that, when $E \geq 2$ and $\varphi(q') \geq 4$, the prime p divides the index exactly in the three families. Two steps decide it, and they are of different kinds: Lemma 13 is a count, how much an interval overlaps its translates, and it makes the residual image full; the moments then make the first layer full. In the stable case the first layer cannot do this — by Theorem 10(d) a full first layer gives $e = 1$ and $v_p = d - 1 > 0$ — which is why the lemma has to come first. Consistently, the 8 stable cells outside the families in $n \leq 25$, $q \leq 60$ all have $q' = 6$, the only value the lemma allows.

Measured: Of 7714 cells outside the families with $n \leq 60$ and $q \leq 150$, the 80 that are stable all have n odd and $q' = r = 6$, and the 26 of those within reach of the instrument are maximal; 195 cells, 39 of them with index > 1 , and no prime outside the families. Proposition 15 against the index summed over all layers: 3060 of 3060 cells with $W_0 = d$, $q \leq 60$, p odd, $E \geq 2$, for $\text{SU}(n)$, $\text{GL}(n)$ and $\text{Sp}(2m)$. And over 147926 unstable segments outside the families with $q \leq 400$ and p odd, none has all its moments divisible by p .

The same count settles the non-real ring of §6, and there the support is narrower.

Corollary 17 (support of the shadow). *For the ring $B_n(q)$ of §6 with $2 \leq n \leq q - 2$: if p divides $[\mathcal{O} : B_n(q)]$, then r divides n or $n - 1$.*

Proof. Here $a_j = j$, $\beta = 1$ and $q' = r$. Suppose r divides neither n nor $n - 1$, so that $B = n \bmod r$ lies in $[2, r - 1]$. A unit fixing the multiset of classes fixes the positions of larger multiplicity, $J = \{0, \dots, B - 1\}$: for $B \leq r - 2$ Lemma 13 leaves $u = \pm 1$, and $u = -1$ sends $1 \in J$ to $r - 1 \notin J$; for $B = r - 1$ it fixes the single class of smaller multiplicity, $r - 1$, so $u = 1$. The stabiliser is trivial, the residual image is full by the argument of Proposition 14, and Nakayama ($E = 1$) or Proposition 15 ($E \geq 2$) applies. For $B \leq r - 2$ the two pairs of the proof of Theorem 16 conclude. For $B = r - 1$ the pair $(0, 1)$ gives $p \mid A + 1$, and then $S_0 + S_{r-1} = A(r - 1) + rA^2 = A(r - 1 + rA) \equiv 1 \pmod{p}$, so not both vanish. $\square$

Measured: Corollary 17 in the only family it excludes that Theorem 16 admits, $r \mid n + 1$: 543 cells with $q \leq 90$, no defect.

5 The two even regimes

For q odd, Corollary 3 has already disposed of n even: the ring is a type- C ring at the same q . What follows is the remaining case, q even, where the complement keeps n even and something has to be said. The prime splits it in two: if $p = 2$ then $v_2(M) = v_2(q) + 1$ and q' is odd, so 2 is a unit; if p is odd then $q' = 2r$ is even and 2 is not.

Proposition 18. *Let $n = 2m$ and let q' be odd. Then multiplication by 2^{-1} carries the segment $\{\pm 1, \pm 3, \dots, \pm(2m - 1)\}$ onto the block of n consecutive residues centred at $q'/2$.*

Proof. Since $2(i - \frac{q'+1}{2}) = 2i - (q' + 1) \equiv 2i - 1$, one has $2^{-1}(2i - 1) \equiv i - \frac{q'+1}{2}$. As i runs from 1 to m these are the m consecutive residues $\frac{q'+1}{2} - m, \dots, \frac{q'+1}{2} - 1$ up to sign, and their negatives are $\frac{q'+1}{2}, \dots, \frac{q'+1}{2} + m - 1$. The union is the block of $2m = n$ consecutive residues from $\frac{q'+1}{2} - m$ to $\frac{q'+1}{2} + m - 1$, whose centre is $\frac{q'+1}{2} - \frac{1}{2} = q'/2$. $\square$

Multiplication by a unit is a Galois automorphism, so it permutes the residue classes; but it does not permute the integer weights $S(c)$ of Theorem 10(b) with them, and the rank can change — at $\text{SU}(4)$, $q = 20$, $p = 2$, $q' = 5$ the moments of the segment are $(0, 1, -3, 3, -1)$, of rank 1 modulo 2, while the shifted sawtooth $(0, 1, 2, -2, -1)$ has rank 2. So the proposition is a statement about classes. Note the difference in strength between this and Corollary 3: a Galois automorphism preserves the rank and the index of whatever it is applied to, the complement preserves the ring itself.

The weights, however, can be computed directly. Write $\hat{s}_f$ for the shifted sawtooth of period f : $\hat{s}_f(0) = \hat{s}_f(f/2) = 0$, $\hat{s}_f(c) = c$ for $0 < c < f/2$ and $c - f$ for $f/2 < c < f$.

Proposition 19 (the first layer in the even regime). *Let p be odd, $v \geq 1$, $r \geq 2$ even with $p \nmid r$, $q = p^v r$, and $n = Nr$ with $1 \leq N < p^v$ — the only stable family with n even. Then $W_1 = 0$ if $p \mid N$, and otherwise W_1 is the rank modulo p of $[\hat{s}_{2r}(u^{-1}c)]$, with rows $u \in (\mathbb{Z}/2r)^\times / \pm 1$ and columns the odd classes c .*

Proof. Here $M = 2q$, $q' = 2r$, and the exponents $a_j = n - 1 - 2j$ are odd. The class c met at position i receives the N exponents $n - 1 - 2i - 2tr$, $0 \leq t < N$, whose sum is $N(r - 1 - 2i)$; and $c \equiv Nr - 1 - 2i \pmod{2r}$. For N odd this reads $S(c) = N \hat{s}_{2r}(c)$; for N even, $S(c) = N(c - r) = N \hat{s}_{2r}((1 + r)c)$, and $1 + r$ is a unit modulo $2r$, so it only permutes the columns. The even classes are not met. Theorem 10(b) now gives the rank, and N is a unit unless $p \mid N$, where every $S(c)$ vanishes. $\square$

What does *not* follow is the identification with the full half-period matrix of [14]. With all columns, $\hat{s}_f(c) = \frac{1}{2} s_f(c + f/2)$ and a translation commutes with the units, so the full shifted matrix has the rank of the ordinary one; but the proposition keeps only the odd columns, and that restriction can lose rank. At $\text{SU}(14)$, $q = 70$, $p = 5$ — so $r = 14$, $N = 1$, and $5 \nmid \varphi(28)$ — the odd columns have rank 5 and all the columns rank 6; the instrument returns $W = (1, 5, 6, 6, \dots)$ and $v_5([\mathcal{O} : A]) = 6$. The loss is not a ladder effect and not a failure of semisimplicity.

Proposition 20. *Let n be even, p odd and $q' = 2r$ with r odd. Then M_{2r} and M_r have the same rank modulo p .*

For q even and p odd the prime-to- p part r of q is even, so this proposition does not apply inside the present section; it covers q odd, where Corollary 3 has already given the ring, and gives the rank there without the complement.

Proof. This is the doubling half of the half-period lemma [14, Lem. 14]: for $r \geq 3$ odd one has $L_{2r} = 2L_r$ between the column lattices, so the elementary divisors differ by a power of 2, which is invisible modulo an odd p . $\square$

Measured: Proposition 18 in 95 of 95 cells and Proposition 20 in 48 of 48. When $q' = 2r$ with r even neither applies — the lemma requires r odd — and there the first layer is given by Proposition 19: its formula for the moments holds in 7580 of 7580 comparisons and its rank in 288 of 288 cells with $r \leq 30$, $p \leq 11$; and the odd columns lose rank against all of them for $r = 14, 28$ with $p = 3, 5, 11$ and for $r = 30$ with $p = 7$.

6 The shadow: the non-real ring of $\mathrm{GL}(n)$

Take instead the element $C_q = \mathrm{diag}(1, \zeta_q, \dots, \zeta_q^{n-1})$ of $\mathrm{GL}(n)$, the principal class of [28]. Its segment $\{0, 1, \dots, n-1\}$ is *not* symmetric, so C_q is not conjugate to its inverse, the values are not real, and $B_n(q)$ — we keep a separate letter for it — is an order in the whole cyclotomic field for $2 \leq n < q$. Indeed σ_u fixes every coefficient exactly when $uI = I$ for $I = \{0, \dots, n-1\} \subset \mathbb{Z}/q$; for $n \leq q-2$ Lemma 13 (with $b = 0$) leaves $u = \pm 1$, and $u = -1$ sends $1 \in I$ to $q-1 \notin I$; for $n = q-1$ the one omitted residue must be fixed, so again $u = 1$. We call it the *shadow* of the real ring.

The restriction $n < q$ is not cosmetic, and Lemma 1 says why: the characteristic polynomial of C_q acquires a factor $T^q - 1$ for each full turn, so

$$B_n(q) = B_{n \bmod q}(q), \quad \text{and in particular } B_n(q) = \mathbb{Z} \text{ when } n \equiv 0, 1 \pmod{q}.$$

Measured: The rank drops to 1 exactly at $n \equiv 0, 1 \pmod{q}$ and is $\varphi(q)$ elsewhere, in 36 cells with $q \leq 7$ and $n \leq 2q+1$.

Lemma 21. $e_k(1, z, \dots, z^{n-1}) = z^{k(k-1)/2} \binom{n}{k}_z$, the Gaussian binomial coefficient twisted by a power of z .

Proof. Immediate from the q -binomial theorem $\prod_{i=0}^{n-1} (1 + z^i t) = \sum_k z^{k(k-1)/2} \binom{n}{k}_z t^k$, comparing coefficients of t^k . $\square$

Measured: 135 checks over $n \leq 7$ and five values of q , with no discrepancy.

That lemma also identifies the residual polynomial used later, and it is not a computation of ours. The q -analogue of Lucas' congruence states, for $n = n_1 d + n_0$ and $k = k_1 d + k_0$ with $0 \leq n_0, k_0 < d$,

$$\binom{n}{k}_q \equiv \binom{n_1}{k_1} \binom{n_0}{k_0}_q \pmod{\Phi_d(q)},$$

a congruence in $\mathbb{Z}[q]$ and not merely an evaluation [18, Thm. 2.2]; the first published proof we know of is Olive's [17], it was rediscovered by Désarménien [16], and Sagan observes it may have been known to Gauss. Reducing modulo the radical sends z to ω of order r , so $d = r$, $n_0 = 0$, and the congruence gives $\binom{Nr}{k}_\omega = \binom{N}{k/r}$ when $r \mid k$ and 0 otherwise. The twisting factor $z^{k(k-1)/2}$ at $k = rs$ becomes $\omega^{rs(rs-1)/2}$, and it is 1 exactly when r divides $rs(rs-1)/2$: for r odd that number is $r \cdot s(rs-1)/2$ with $s(rs-1)$ always even, so the factor is 1; for r even it is $(r/2)s(rs-1)$ with $rs-1$ odd, divisible by r only when s is even, and for s odd the exponent is $r/2$ modulo r and the factor is -1 . So

$$e_{rs} \equiv (-1)^{s(r+1)} \binom{N}{s} \pmod{\mathfrak{m}}, \quad \prod_{j < Nr} (T - \omega^j) = (T^r - 1)^N,$$

the sign being $+1$ throughout the odd r of this note, and the second identity holding in both parities — and it is the residual polynomial that Theorem 25 runs on, so the sign does not reach the bridge.

Measured: The coefficients e_{rs} computed exactly in $\mathbb{Z}[\zeta_r]$ over $r = 3, \dots, 13$ in *both* parities and $N \leq 4$, that is 44 pairs (r, N) : the signed formula above matches all 154 coefficients e_{rs} , and the unsigned one $\binom{N}{s}$ fails in 30 — all of them with r even and s odd, the smallest being $r = 4, N = 1$, where $e_4 = -1$. The second identity, $\prod_{j < Nr} (T - \omega^j) = (T^r - 1)^N$, holds in all 44 pairs, even r included, which is why Theorem 25 is untouched by the sign.

So the shadow is the ring generated by twisted Gaussian binomials evaluated at a root of unity. Its local picture is the same theorem with two changes, both caused by the loss of the symmetry.

Theorem 22 (the shadow). *Assume the segment $\{0, \dots, n-1\}$ is stable at q' and $E \geq 2$, and put $D = \varphi(q')$. Then W_1 is the rank modulo p of $[S(u^{-1}c)]$ with rows over all of $(\mathbb{Z}/q')^\times$ and columns over all classes, and*

$$v_p([\mathcal{O} : B_n]) = \sum_{i < e} (D - W_i), \quad e = 2 \implies v_p = 2D - 1 - W_1.$$

Proof. Lemma 6 does not use the symmetry of the segment, so it applies unchanged — and here it is used as it stands, without the passage to real coordinates of Lemma 7, which is exactly what is not available. What changes is the basis: here one may take $\{\gamma_c F(T)/(T - \gamma_c)\}$ over *all* classes, which is independent and satisfies $\sigma_u(\gamma_c F/(T - \gamma_c)) = \gamma_{uc} F/(T - \gamma_{uc})$ by the stability of F ; the row of u is again $(S(u^{-1}c))_c$. The quantum integers U_c of Lemma 7 were exactly what implemented the descent to the real subring, and are not needed now. Consequently no class is discarded, the rows are not identified in pairs, and each graded layer of $\mathcal{O}_p$ has dimension $D = \varphi(q')$ rather than $\varphi(q')/2$. Parts (c) and (d) are unchanged. $\square$

Measured: Theorem 22 in 75 of 75 cells under $W_1 > D/2$, with no discrepancy, and the description of W_1 in 81 of 81. The large shadow indices that no closed formula in (n, q) could reproduce — 7^4 at $(n, q) = (5, 35)$, 5^7 at $(7, 35)$ and at $(9, 45)$, 3^7 at $(8, 21)$ — are this law with D integral. The index 2^7 at $(7, 14)$ has $E = \varphi(2) = 1$ and is *outside* the hypothesis of the theorem; it is recorded, not explained.

Proposition 23 (the first layer of the shadow is one trivial direction plus Stickelberger). *Let p be odd, $v \geq 1$, and let $r \geq 3$ be coprime to p — of either parity. Put $q = p^v r$ and let $n = Nr$ or $n = Nr + 1$ with $N \geq 1$ and $2 \leq n < q$. Then the image of $B_n(q)$ in $\mathcal{O}_p/\mathfrak{r}$ is $\mathbb{F}_p$, and*

$$p \nmid N : W_1(B_n(q)) = 1 + \text{rank}_{\mathbb{F}_p} \mathcal{M}_r; \quad p \mid N : W_1(B_n(q)) = 0.$$

Here $\mathcal{M}_r$ is the moment matrix of [14], built from the sawtooth $s_r(0) = 0$, $s_r(c) = 2c - r$, with rows $u \in (\mathbb{Z}/r)^\times / \pm 1$ and columns $1 \leq c < r$. No condition on its rank is needed, and none on whether p divides $\varphi(r)$.

In particular the reference rank of the shadow's first layer is $d + 1$ with $d = \varphi(r)/2$, and *not* $D = 2d$: the nontrivial even characters are absent before any divisibility of h^- enters, and the trivial one is the extra direction. Note also what is *not* assumed. The hypothesis $E \geq 2$ of Theorem 22 is not an extra condition here but arithmetic, $E = p^{v-1}(p-1) \geq 2$ for every odd p ; and r need not be odd, which for the real ring it must be. This is the first statement of the note that holds on the shadow under weaker hypotheses than on the real side.

Proof. The first-order quotient. Write $\zeta_q = \omega\rho$ with ω of order r and ρ of order p^v , and put $\pi = \rho - 1$, $R_0 = \mathbb{Z}_p[\omega]$. Since $p \nmid r$, the map $R_0[Y]/(\Phi_{p^v}(1+Y)) \rightarrow \mathcal{O}_p$, $Y \mapsto \pi$, is onto — $\mathcal{O}_p = R_0[1+\pi]$ — between free $\mathbb{Z}_p$ -modules of the same rank $\varphi(r)E$, hence an isomorphism. $\Phi_{p^v}(1+Y)$ is Eisenstein over each factor of R_0 : it reduces to Y^E and its constant term is $\Phi_{p^v}(1) = p$. So $\mathfrak{r} = \pi\mathcal{O}_p$ and $k := \mathcal{O}_p/\mathfrak{r} \cong \mathbb{F}_p[X]/(\bar{\Phi}_r)$ is étale of dimension $\varphi(r)$ — a product of fields, not a field. Modulo Y^2 the Eisenstein relation reads $p + c_1 Y = 0$ with $p \mid c_1$, *because* $E \geq 2$ makes c_1 a non-leading coefficient; so it reads $p(1 + aY) = 0$ with $1 + aY$ a unit, that is $p = 0$, and

$$\Lambda := \mathcal{O}_p/\mathfrak{r}^2 \cong k[\varepsilon]/(\varepsilon^2), \quad \pi \mapsto \varepsilon,$$

with the image of $\mathbb{Z}_p$ exactly $\mathbb{F}_p$. This is the one place $E \geq 2$ is used: at $E = 1$ the integer p would itself contribute a first-order direction.

The image is $\mathbb{F}_p \oplus \varepsilon V$. In Λ , $\zeta_q^a = w^a(1 + a\varepsilon)$, so expanding and discarding ε^2 ,

$$Q(T) = \prod_{a < n} (T - \zeta_q^a) = F(T) - \varepsilon H(T), \quad H(T) = \sum_{c \in \mathbb{Z}/r} S(c) w^c \frac{F(T)}{T - w^c},$$

with $F = (T^r - 1)^N$ or $(T - 1)(T^r - 1)^N$ — in $\mathbb{F}_p[T]$, which is what the two families buy — and $S(c) = \sum_{a < n, a \equiv c} a$. Grouping by class is legitimate because deleting one factor of the class c leaves $F/(T - w^c)$ whichever a it was. The generators of the image are the coefficients $f_j - \varepsilon h_j$ with $f_j \in \mathbb{F}_p$; writing V for the $\mathbb{F}_p$ -span of the h_j , the set $\mathbb{F}_p \oplus \varepsilon V$ contains them, contains 1, and is closed under multiplication, since $(\alpha + \varepsilon u)(\beta + \varepsilon v) = \alpha\beta + \varepsilon(\alpha v + \beta u)$ and the scalars are $\mathbb{F}_p$; so the image Γ is contained in it. Conversely Γ contains $\mathbb{F}_p$, hence each $\varepsilon h_j = f_j - (f_j - \varepsilon h_j)$, hence εV . So $\Gamma = \mathbb{F}_p \oplus \varepsilon V$, the residual image is $\mathbb{F}_p$, and from the exact sequence $0 \rightarrow B \cap \mathfrak{r}^2 \rightarrow B \cap \mathfrak{r} \rightarrow \varepsilon V \rightarrow 0$ one reads $W_1 = \dim V$. It is the scalar part being $\mathbb{F}_p$ that closes this, not $\varepsilon^2 = 0$ alone.

$\dim V$ is the rank of the full moment matrix. Extend scalars to $\overline{\mathbb{F}}_p$ and let $\sigma_u(w) = \xi^u$ run over the $\varphi(r)$ embeddings. Then $\sigma_u(H) = \sum_c S(u^{-1}c)P_c$ with $P_c = \xi^c F/(T - \xi^c)$, and the P_c are independent: from $\sum_c \lambda_c P_c = 0$, cancel F in $\overline{\mathbb{F}}_p(T)$, multiply by $T^r - 1$ and evaluate at ξ^b , where every term but b dies and the survivor is $r\lambda_b$, so $\lambda_b = 0$ as $p \nmid r$. The cancellation is essential: the roots of F may have multiplicity divisible by p , so evaluating the P_c at them directly would prove nothing. Hence $\dim V$ is the rank of $[\overline{S}(u^{-1}c)]$ over all u and all c .

The parity split. Counting the arithmetic progressions gives, as integers, $S(c) = Nc + \frac{rN(N-1)}{2}$, plus Nr at $c = 0$ when $n = Nr + 1$. With $j = -1 \in (\mathbb{Z}/r)^\times$ and p odd, $e_\pm = (1 \pm j)/2$ are central idempotents with $e_\pm \overline{S} = S^\pm$, so $\mathbb{F}_p[G]\overline{S} = \mathbb{F}_p[G]S^+ \oplus \mathbb{F}_p[G]S^-$. No semisimplicity is used — only that 2 is invertible — so this survives $p \mid \varphi(r)$. One computes $S^- = \frac{N}{2} s_r$, while S^+ takes the single value $rN^2/2$ on every nonzero class and so is fixed by every unit. If $p \nmid N$ then $S^+ \neq 0$ and contributes exactly one dimension, and $\frac{N}{2} s_r$ generates what s_r does: deleting the zero column and keeping one row of each pair $\pm u$ leaves $\mathcal{M}_r$. If $p \mid N$ then $\overline{S} \equiv 0$ identically, $H = 0$, and every coefficient of Q is a scalar: $B \cap \mathfrak{r} = B \cap \mathfrak{r}^2$ and $W_1 = 0$. $\square$

Measured: 100 cells by the slow path — closing the lattice in degree $\varphi(q)$ and not through the bridge, which would be circular — over $q = 45, 63, 99, 117, 207, 35, 55$, with the twenty $p \mid N$ cells returning 0 where $1 + \text{rank}$ would have said 3, 4, 6, 7 or 11. The formula is distinguishable from the trivial $1 + d$ only where the rank drops, so the witnesses are the point: $r = 23$ with $p = 3$ gives defect 1, and so does $r = 31$ with $p = 3$ in the profile at $q = 279$ that `autotest` reproduces; $r = 41$ with $p = 11$ — at $q = 451$, $v = 1$ — gives defect 2, with $W_1 = 19$ against $1 + d = 21$. The relaxations are measured too: $r = 3$ at $q = 15, 33, 75$ and $r = 4$ at $q = 36, 100$.

The three hypotheses that remain are not decoration, and each has a witness. At $p = 2$ the split fails: $q = 20$, $n = 5$ has $W_1 = 3$ while $1 + \text{rank}_{\mathbb{F}_2} \mathcal{M}_5 = 2$. If $p \mid r$ the residual algebra is not étale and the statement is not even well posed: reading $q = 27$ with $r = 9$ gives $W_1 = 0$ against a predicted 2. And outside the two families the residual image exceeds $\mathbb{F}_p$, which is what the proof's closure step needs: $B_2(15) = \mathbb{Z}[\zeta_{15}]$ is maximal, with $W_1 = \varphi(5) = 4$ against 3.

Observation 24 (the class number is visible in the shadow too). On the segments of length $Nr \pm 1$, where the Euler factor at 2 does not enter, the drop of the first layer follows the class number on both sides. At $r = 41$ with $p = 11 \mid h^- = 121$ the real first layer $W_1(R)$ and the shadow's drop by exactly 2 (at $q = 451$); at $r = 23$ with $p = 3 \mid h^- = 3$, by one (at $q = 207$). With $p \nmid h^-$ they are full: at $r = 23$ with $p = 5$ ($q = 115$) and at $r = 17, 19$, where $h^- = 1$. On the segments that cover whole periods a second source adds to this one, the factor $\chi(2) - 1$: at $q' = 31$ it removes $d/t = 3$ dimensions with $h^-(\mathbb{Q}(\zeta_{31})) = 9$ prime to the p in question [32].

The shadow is a quadratic extension of a real ring already computed

Proposition 23 and the observation above say that something happens in the shadow — the first layer is $d + 1$ and not $2d$, and the class number is visible — without saying what the shadow is. There is an answer, and it is a statement about rings rather than about dimensions: the non-real ring is not a separate regime but a free quadratic extension of the ring of a segment of the leftmost position of Figure 2. The proof is Lemma 1 used backwards. The segment $\{0, \dots, n-1\}$ is not centred anywhere useful, but the division lemma moves it, without changing the ring, to a representative that is symmetric about $Nr/2$ — and the element that recentres that one is a root of unity of p -power order, with no component along ω . That is what the ring turns out to contain.

One condition is about to be used under two names, and they are the same condition. For the segment $\{0, \dots, n-1\}$ the lengths $n = Nr$ and $n = Nr + 1$ are those for which the residue counts N_c are constant off the class 0 — stability in the sense of Definition 8, by Lemma 9 — and they are also the lengths for which the division lemma moves the segment to a representative symmetric about $Nr/2$, whose recentring element $\zeta_q^{Nr/2}$ has no component along ω and lives entirely in the ramified direction. The first description is what §3 needs, and the second is what makes the theorem below work. It is not a criterion for a defect in general: a symmetric segment is centred at 0 already.

Write $\mathcal{O}_+$ for the real subring of $\mathcal{O}_p$, e_1 for the first elementary symmetric function of the eigenvalues and $\bar{e}_1$ for its complex conjugate.

Theorem 25 (the bridge). *Let p be odd, $r \geq 5$ odd with $p \nmid r$, $q = p^v r$, and $n = Nr$ or $n = Nr + 1$ with $2 \leq n < q$; put $P = p^{v_p(N)}$, so that $P < p^v$. Write $B = B_n(q) \otimes \mathbb{Z}_p$, $R = B \cap \mathcal{O}_+$, and $\theta = e_1 - \bar{e}_1$. Let S be the segment $\{1, \dots, Nr-1\}$ if $n = Nr$ and $\{0, \dots, Nr\}$ if $n = Nr + 1$, let $\mu = \zeta_q^{Nr/2}$, and let A be the ring generated by the character values of the centred segment $S - Nr/2$. Then*

$$\mu \in B, \quad B = A[\mu], \quad B = R \oplus \theta R,$$

free of rank 2 with $\theta^2 \in R$, and θ is anti-invariant of valuation exactly — on elements of $\mathcal{O}_p$ we normalise $v_p(\pi) = 1$, so that $v_p(p) = E$, while on indices v_p is the ordinary valuation of an integer —

$$v_p(\theta) = v_p(\zeta_q^{Nr} - 1) = P.$$

Consequently, writing $H(t) = \sum_i W_i t^i$ for the Hilbert series of the layers,

$$H_B(t) = (1 + t^P) H_R(t), \quad e_B = e_R + P, \quad v_p([\mathcal{O} : B_n]) = 2 v_p([\mathcal{O}_+ : R]) + P d.$$

Proof. Since q is odd, 2 is invertible modulo q and $\mu = \zeta_q^{Nr \cdot 2^{-1}}$ is defined; writing $\zeta_q = \omega \rho$ as in §3, $\omega^{Nr} = 1$ forces $\mu = \rho^{Nr \cdot 2^{-1}}$, a root of unity of p -power order with no ω -part.

The representative. When $n = Nr$ the exponent 0 lies in the segment, so $\prod_{j < n} (T - \zeta_q^j) = (T-1)g(T)$ with $T-1$ monic in $\mathbb{Z}[T]$; by Lemma 1 the ring is unchanged and g is the polynomial of $S = \{1, \dots, Nr-1\}$. When $n = Nr + 1$ take S as it stands. In both cases S is symmetric about $Nr/2$, so $e_k(S) = \mu^k a_k$ with $a_k = e_k(\zeta_q^c : c \in S - Nr/2)$ real, and $A = \mathbb{Z}_p[a_1, \dots]$.

$e_1(S)$ is a unit of B . By Proposition 23 the image of B in $\mathcal{O}_p/\mathfrak{t}$ is $\mathbb{F}_p$, so B , being finite over $\mathbb{Z}_p$, is local with residue field $\mathbb{F}_p$, and e_k reduces to the corresponding coefficient of the residual polynomial. For $n = Nr + 1$ that polynomial is $(T-1)(T^r-1)^N$ and $e_1(S) \equiv 1$; for $n = Nr$ one has $e_1(S) = e_1 - 1 \equiv -1$, because $\sum_{j < Nr} \omega^j = N \sum_c \omega^c = 0$. Either way the residue is $\pm 1 \neq 0$. Note that this zero comes from summing over *all* the classes and not from N : the step does not care whether p divides N , and neither does any other.

$\mu \in B$. As a_1 is real, $\bar{e}_1(S) = \mu^{-1}a_1$, so $\mu^2 = e_1(S)/\bar{e}_1(S)$, a quotient by a unit of B , hence in B ; and μ has odd order, so $\mu = (\mu^2)^{(m+1)/2}$ with m the order of μ^2 . Hence $a_k = \mu^{-k}e_k(S) \in B$, so $A \subseteq B$ and $B = A[\mu]$. (In particular $\rho^P \in B$ and $B_{Nr}(q) \otimes \mathbb{Z}_p = B_{Nr+1}(q) \otimes \mathbb{Z}_p$. Not ρ itself: by the valuation step below $\mu^2 = \rho^{Nr}$ has order $p^{v-v_p(N)}$, so $\langle \mu \rangle = \langle \rho^P \rangle$, which is a proper subgroup exactly when $p \mid N$. It has to be proper there, because Proposition 23 gives $W_1(B) = 0$ when $p \mid N$, and $\rho \in B$ would put $\pi = \rho - 1$, of valuation 1, into B .)

The extension. Put $\tau = \mu + \mu^{-1}$, fixed by conjugation. From $\mu^2 - \tau\mu + 1 = 0$ the module $B = R[\mu]$ is generated over R by $1, \mu$, and freely: if $a + b\mu = 0$ with a, b real then conjugating and subtracting gives $b(\mu - \mu^{-1}) = 0$, so $b = a = 0$. An element $a + b\mu$ is anti-invariant exactly when $a = -b\tau/2$, so $B^- = (\mu - \mu^{-1})R$. Finally θ is unchanged by the passage to S — for $n = Nr$ the constant cancels in $e_1 - \bar{e}_1$ — and $\theta = a_1(\mu - \mu^{-1})$ with $a_1 = e_1(S)/\mu$ a unit, so $\theta R = (\mu - \mu^{-1})R = B^-$.

The valuation. $\mu - \mu^{-1} = \mu^{-1}(\mu^2 - 1)$ and $\mu^2 = \zeta_q^{Nr} = \rho^{Nr}$, a root of unity of order $p^{v-v_p(N)}$ because $p \nmid r$; a root of unity of order p^{v-a} is $1 +$ (an element of valuation p^a), so $v_p(\theta) = p^{v_p(N)} = P$. Freeness needs $\mu^2 \neq 1$, that is $P < p^v$, and that is automatic: $B \neq \mathbb{Z}$ forces $Nr < q$ by Lemma 1, exactly as in [14, Prop. 19]. Finally the filtration. As p is odd, the projectors $(1 \pm \iota)/2$, with ι the conjugation, preserve every power of $\mathfrak{r}$, and $\mathcal{O}_p$ is unramified over $\mathcal{O}_+$ because $r \geq 3$; so $\mathfrak{r}^i \cap \mathcal{O}_+ = \mathfrak{r}_+^i$ and, θ having valuation exactly P ,

$$B \cap \mathfrak{r}^i = (R \cap \mathfrak{r}_+^i) \oplus \theta(R \cap \mathfrak{r}_+^{i-P}),$$

reading the second term as θR when $i < P$. Hence $\text{gr}(B) = \text{gr}(R) \oplus \bar{\theta} \text{gr}(R)$ with the second summand in degrees shifted by P ; that is the Hilbert series identity, and evaluating $\sum_i (2d - W_i(B))$ gives the index. $\square$

The theorem does not say that R is A ; it says $A \subseteq R$ and that B is free of rank 2 over R . Since $B = A[\mu] = A[\tau][\mu]$ and B is free of rank 2 over $A[\tau]$ by the same computation, $R = A[\tau]$ exactly, and the question is whether the single element $\tau = \mu + \mu^{-1}$ already lies in A . It does, and for a reason that costs no hypothesis at all: τ is a rational function of the first two coefficients, with a denominator that is a unit.

Proposition 26. *Under the hypotheses of Theorem 25, $\tau \in A$ and therefore $R = A$. Writing L for the length of the centred representative and a_1, a_2 for the first two elementary symmetric functions of its spectrum,*

$$L = Nr + 1 : \quad \tau = \frac{a_1(1 + a_1^2 - 2a_2)}{a_1^2 - a_2}; \quad L = Nr - 1 : \quad \tau = \frac{a_1(a_1^2 - 2a_2 - 1)}{a_2}.$$

No condition on the conductor exponent, on u , or on $p \nmid N$ is needed.

Proof. Both displays are identities in $\mathbb{Z}[t, t^{-1}]$ for the spectrum $\{t^{L-1}, t^{L-3}, \dots, t^{1-L}\}$ with $\tau = t^{Nr} + t^{-Nr}$, so they survive every specialisation of t — in particular $t \mapsto \zeta_q^{(q+1)/2}$, which turns the spectrum into that of $S - Nr/2$ and t^{Nr} into μ (with $t \mapsto \zeta_{2q}$ instead, for N odd, every eigenvalue and τ would change sign); they follow from $[L+1]_t + [L-1]_t = [2]_t[L]_t$ and $[L]_t^2 - [L-1]_t[L+1]_t = 1$ together with $a_1 = [L]_t$ and $a_2 = [L]_t[L-1]_t/[2]_t$. It remains that the denominator used in each case is a unit, and this is where the two families part. The residual polynomial of the centred representative is $(T-1)(T^r-1)^N$ when $L = Nr+1$ and $(T^r-1)^N/(T-1)$ when $L = Nr-1$ — recentering does not move the residues, μ having p -power order — so $(\bar{a}_1, \bar{a}_2) = (1, 0)$ in the first case and $(-1, 1)$ in the second. Hence $\bar{a}_1^2 - \bar{a}_2 = 1$ in the first and $\bar{a}_2 = 1$ in the second, each the denominator that appears in its own formula. The reduction is modulo $\mathfrak{r}_+$, not modulo $p\mathcal{O}_+$, and that is all that is needed: a unit is a unit. Since $a_1, a_2 \in A$, so is τ , and $R = A[\tau] = A$. $\square$

The two denominators are never units together — $\bar{a}_2 = 0$ in the first family and $\overline{a_1^2 - a_2} = 0$ in the second — so the two formulas are not interchangeable, and neither is a special case of the other.

Measured: The two identities hold in $\mathbb{Z}[t, t^{-1}]$ for every length $2 \leq L \leq 40, 78$ of 78, and the residues $(1, 0)$ and $(-1, 1)$ come out in 70 of 70 cells over $p \in \{3, 5, 7\}$, $r \in \{5, 7, 11, 13\}$ and $N \leq 7$ — the fourteen with $p \mid N$ included, where nothing changes. The specialisation is checked separately: 68 exact identities in $\mathbb{Z}[\zeta_q]$ over $q = 45, 63, 99, 117, 35, 55, 175$.

With $R = A$, the real side is computed by [14]. Write $u = \text{rank}_{\mathbb{F}_p} \mathcal{M}_r$ for the rank of the moment matrix (a number, not a unit of $\mathbb{Z}/q'$). For $p \nmid N$, [14] gives $W_1(R) = u$, and when moreover $u > d/2$ — which Proposition 26 no longer needs, but the profile of [14] still does — it gives $e_R \leq 2$, so with $\delta = d - u$ and $P = 1$ the bridge reads

$$W(B) = (1, 1 + u, d + u, 2d, 2d, \dots), \quad e_B \leq 3, \quad v_p([\mathcal{O} : B_n]) = 3d - 2 + 2\delta,$$

and the shadow's conductor is the cube of the radical exactly where the real one is its square. Figure 5 is that sentence drawn.

The second entry of that profile is worth recognising: $W_1(B) = 1 + u$ is Proposition 23, arrived at here as a corollary of the bridge rather than proved directly. The two routes are independent — the proposition never mentions R , and the bridge never counts moments — and they agree, which is the only reason the coincidence is worth printing. They do not have the same reach: the bridge needs r odd and at least 5, and returns the whole profile; the proposition needs neither, and returns only the first layer. Where both apply, the bridge *explains* the shape the proposition computes — the trivial direction is $W_0(R) = 1$ shifted by $P = 1$, and Stickelberger's rank is $W_1(R) = u$. Where the square happens is the subject of [14, Thm. 12]: in this family δ counts the invariant factors of $\mathcal{M}_r$ divisible by p , and a simple divisor of $h^-(\mathbb{Q}(\zeta_r))$ gives $\delta = 1$ [14, Cor. 16]. The first conductor at which it happens is $r = 23$ with $p = 3$, where $h^- = 3$.

When $p \mid N$ the same P is already in [14]: below $p^{v_p(N)}$ the real layers there are a universal ladder, $W_i = 1$ for $i \in \{P - p^j\}$ and 0 otherwise [14, Prop. 19]. By Theorem 27 below R follows it, so the shift of the bridge and the length of the ladder are the same number, and the ladder supplies the first term of $v_p([\mathcal{O} : B_n]) = 2v_p([\mathcal{O}_+ : R]) + P d$. The non-degeneracy $P < p^v$ is not an extra hypothesis either: it is the same $v > a$ that [14, Prop. 19] derives from $A \neq \mathbb{Z}$.

[14, Prop. 19] assumes a segment of length $2m + 1 = Nq'$, and the centred segment here has length $Nr \pm 1$; the ladder transfers all the same.

Theorem 27 (the ladder at the centred segment). *Under the hypotheses of Theorem 25, write $a = v_p(N)$ and $P = p^a$. For both centred representatives, of lengths $Nr - 1$ and $Nr + 1$, and for $0 \leq i < P$,*

$$W_i(R) = 1 \quad \text{if } i \in \{P - p^j : 0 \leq j \leq a\}, \quad W_i(R) = 0 \quad \text{otherwise.}$$

Proof. Work modulo $\mathfrak{r}^P$. Since $N < p^v$, $a < v$ and $E = (p - 1)p^{v-1} \geq P$, so $p \equiv 0$ there. The recentring element satisfies $\mu^2 = \rho^{Nr}$ with $v(\mu^2 - 1) = P$, and μ has odd order, so $\mu \equiv 1$: recentring changes no coefficient at this depth, and by Lemma 1 neither does the factor $T - 1$. So the image of R is the ring generated by the coefficients of $Q(T) = \prod_{j < Nr} (T - \zeta_q^j)$. Write $N = Ph$ with $p \nmid h$ and $t = \rho$, so that $t^P \equiv 1$ because $v(\rho^P - 1) = P$. As j runs over h full periods of length rP , the pair $(j \bmod r, j \bmod P)$ takes every value h times, and $\prod_c (T - \omega^c y) = T^r - y^r$; hence $Q \equiv F_P(T^r)^h$ with $F_P(X) = \prod_{b < P} (X - t^b)$, and since $p \nmid h$ the coefficients of F_P are polynomials in those of F_P^h (an h -th root of a monic series). Everything happens in $C = \mathbb{F}_p[t]/(t^P - 1) = \mathbb{F}_p[\pi]/(\pi^P)$. The coefficients of F_P are fixed by $t \mapsto t^u$ for every unit u modulo P , so they lie in C^U , whose basis of orbit sums has $a + 1$ elements; since

$\sum_{\ell < P/p^j} t^{\ell p^j} = (t^P - 1)/(t^{p^j} - 1) = \pi^{P-p^j}$, a basis of C^U is 1 and π^{P-p^j} , $0 \leq j < a$. On the other side, $e_k(1, t, \dots, t^{P-1}) = t^{k(k-1)/2} \binom{P}{k}_t$ and $v_\pi(1 - t^m) = p^{v_p(m)}$ in characteristic p , so

$$v_\pi(e_k) = \sum_{j=0}^{k-1} p^{v_p(P-j)} - \sum_{j=1}^k p^{v_p(j)} = P - p^{v_p(k)},$$

using $v_p(P - j) = v_p(j)$ for $0 < j < P$. The coefficients e_{p^j} , $0 \leq j < a$, have the distinct valuations $P - p^j$, and with 1 they span C^U . So the image of R modulo $\mathfrak{r}^P$ is C^U , with one dimension in each degree $P - p^j$ and none elsewhere. $\square$

Measured: Both centred representatives at $q = 135$, $p = 3$, $N = 9$, of lengths 44 and 46: $(W_0, \dots, W_8) = (1, 0, 0, 0, 0, 0, 1, 0, 1)$, as the theorem says, and at $q = 45$, $p = 3$, $N = 3$ the profile $(1, 0, 1, 2, 2, 2)$ of Figure 5(c).

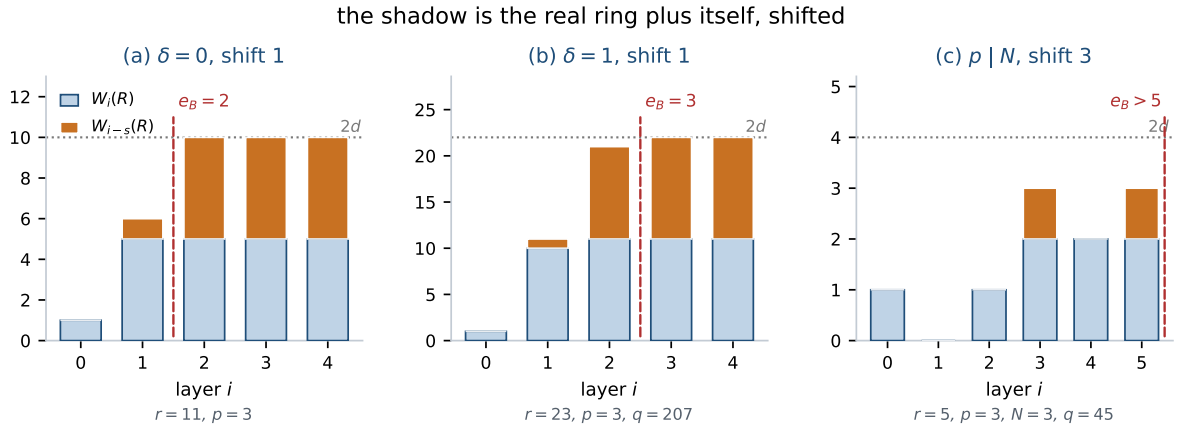


Figure 5: Theorem 25 as an arithmetic of bars: the pale bar is $W_i(R)$, the dark one is the same bar moved P places to the right, and the total height is $W_i(B)$. The dotted line is the full dimension $2d$; the conductor exponent of the shadow is the first layer from which the two bars together reach it, which is P more than for R alone — that is $e_B = e_R + P$, seen rather than believed. (a) the generic case. (b) the first conductor where h^- bites, $r = 23$ at $p = 3$: the real ring loses one dimension in layer 1, the shadow loses it twice, one layer apart, and the conductor goes from the square of the radical to its cube. (c) when $p \mid N$ the shift is $P = p^{v_p(N)}$ and the pale bar has the ladder shape of Theorem 27; here the window stops at $K = 6$ before the bars fill, so the mark is a bound and not a value.

Proposition 28 (the type crosses the bridge). *Under the hypotheses of Theorem 25, $\text{type}(B) = \text{type}(R)$. If moreover $p \nmid N$ and $d/2 < u < d$, then $\text{type}(B) = \delta + \kappa - 1$ with κ as in [14, Prop. 11]; in particular B is Gorenstein exactly when $\delta = \kappa = 1$, although its conductor is one power ($P = 1$) deeper than that of R .*

Proof. B is free, hence flat, over the local ring R , and θ^2 lies in the maximal ideal of R because it has valuation $2P \geq 2$, so the closed fibre is $\mathbb{F}_p[T]/(T^2)$, which is Gorenstein of type 1. For a flat local homomorphism with Gorenstein closed fibre the canonical module is the base change of the canonical module, so the type is multiplied by that of the fibre [25, §3.3]; the residue fields agree. The second statement is [14, Prop. 11] applied to R , which has $e_R = 2$ there: with $p \nmid N$ one has $W_1(R) = u$, and $d/2 < u < d$ is the hypothesis of that proposition. $\square$

A conductor that is the square of the radical does not make R Gorenstein. With $\ell(\mathcal{O}_+/R) = \sum_{i < e} (d - W_i)$ and $\ell(R/\mathfrak{f}_R) = \sum_{i < e} W_i$, the Gorenstein condition $\ell(\mathcal{O}_+/R) = \ell(R/\mathfrak{f}_R)$ reads $\sum_{i < e} W_i = ed/2$, which is a separate condition.

Measured: At $r = 41$, $p = 11$, $q = 451$, $N = 1$ the real profile is $(1, 18, 20, 20)$, so $e_R = 2$ and the real conductor is the square of the radical, while $\ell(\mathcal{O}_+/R) = 21 \neq 19 = \ell(R/\mathfrak{f}_R)$: R is not Gorenstein, and by Proposition 28 neither is B , whose conductor is the cube.

	$W(R)$	$W(B)$	$e_R \rightarrow e_B$	$v_p([\mathcal{O}_+ : R])$	$v_p([\mathcal{O} : B])$
$\delta = 0$	$(1, d, d, \dots)$	$(1, d+1, 2d, 2d, \dots)$	$1 \rightarrow 2$	$d - 1$	$3d - 2$
$\delta > 0$	$(1, d-\delta, d, \dots)$	$(1, d+1-\delta, 2d-\delta, 2d, \dots)$	$2 \rightarrow 3$	$2d - 1 - u$	$3d - 2 + 2\delta$
$q = 207, p = 3$	$(1, 10, 11, \dots)$	$(1, 11, 21, 22, \dots)$	$2 \rightarrow 3$	11	33
$q = 279, p = 3$	$(1, 14, 15, \dots)$	$(1, 15, 29, 30, \dots)$	$2 \rightarrow 3$	$15^\dagger$	$45^\dagger$

The bridge in two regimes and in the only two cells of the swept range with $\delta > 0$, both r prime with $p = 3 \mid h^-$ and $\delta = 1$. The layers are measured; the two indices at $q = 207$ are closed integer lattices, and the two marked $\dagger$ are the law applied to the measured layers, not lattices. At $q = 207$ the plane of §3 would give 32.

Measured: Independent checks: the displayed identities hold in 306 of 306 cells read to depth $K = \min(E, 4)$, the index law in 13 of 13 cells by closing integer lattices rather than layers, and the seven steps of the proof — the representative, the unit, $\mu \in B$, $B = A[\mu]$, $R = A$, $B^- = \theta R$ and $v_p(\theta) = P$ — each separately in 66 of 66 cells with integer lattices localised at p , at $P = 1, 3$ and 9 , with $\varphi(p^v) = 2$, with composite r and with $N \leq 9$. Each of the seven is covered by the proof.

Three remarks on what the proof does *not* use. It does not need $E \geq 3$: nothing is expanded to second order in π . Neither it nor Proposition 26 needs $u > d/2$; that hypothesis enters only through what [14] can say about the profile of the real ring, which is what pins $e_R \leq 2$ and hence the display. And it does not name a symplectic ring: R is the ring of the centred segment, and the identification of that ring with $A_{\mathrm{Sp}(2m)}(q)$ for a suitable m is Proposition 2 and Corollary 3 applied to $S - Nr/2$, whose length is $Nr \pm 1$ — so m depends on N , and the naive guess $m = r - 1$ is wrong even locally: at $q = 45$, $p = 3$, $N = 3$ the ring R has $v_3([\mathcal{O}_+ : R]) = 4$ and $W_1(R) = 0$, while $A_{\mathrm{Sp}(8)}(45)$ has $v_3 = 1$ and $W_1 = 2$. Over 9 cells and 54 comparisons $A_{\mathrm{Sp}(2m)}(q)$ with $m = r - 1$, $m = r$ and $m = (q - r - 2)/2$ give the same truncated image while $m = r - 2$ and $m = r + 1$ do not, and $m = r - 3$ does only in the two cells at $q = 45$, and the integer lattices of the three are already distinct at $q = 45$, of indices 75, 3 and 3.

What the shift measures is the degree of the anti-invariant generator, and that degree has a reading of its own. Write n' for whichever of n , $n - 1$ is divisible by r ; then $P = v_p(\zeta_q^{n'} - 1)$ is how far the segment is from closing into a full period. When it does close, $\zeta_q^{n'} = 1$ and the extension degenerates — which is the other thing this section already knew, $B_n(q) = \mathbb{Z}$ at $n \equiv 0, 1 \pmod{q}$. One number governs both ends.

The shape of Theorem 25 is not new: whether an order in a CM field is free of rank two over the subring fixed by complex conjugation is a hypothesis Howe isolated and used [8, pp. 13, 20], and, in the quartic CM setting of [10], when the real ring is the *maximal* one the conductor of such an extension is computed and the order is Gorenstein because it is monogenic [10, Lemmas 7, 8] — a precedent of shape, not a proof for cyclotomic orders of arbitrary degree. The graded consequence has a standard shape too: for Artinian graded algebras, $\mathrm{gr}(B)$ over $\mathrm{gr}(R)$ with fibre $\mathbb{F}_p[\bar{\theta}]/(\bar{\theta}^2)$ is a *free extension* in the sense of Harima and Watanabe, whose Hilbert function multiplies [9, Def. 1.5, Lem. 1.6]. Our graded rings are not Artinian, so we cite this as context: here the product of Hilbert series is proved by the filtered direct sum in the proof above, not by [9]. What is new is which orders these are, and that the real one is not maximal: there the layers of R are not all full and the convolution has content.

▷ **Where it does break.** *Not at $p \mid N$: nothing in the proof looks at N modulo p , and the only thing that changes there is the value of P . It breaks at $p = 2$: the modulus q is even and 2^{-1} does not exist modulo q . The centring element may still exist when Nr is even — $q = 20$, $r = 5$, $N = 2$ has $\mu = \zeta_{20}^5$ —*

but the proof recovers μ from μ^2 through its odd order and splits B with the projectors $(1 \pm \iota)/2$, and both steps divide by 2. That is a different problem, not a harder case of this one — the index 2^7 at $(n, q) = (7, 14)$, recorded and unexplained after Theorem 22, is its smallest instance. For $u \leq d/2$ the ring is still A , by Proposition 26, but [14] does not give the profile of A there, so the displayed $W(B)$ does not follow. That is a gap in the real input, not in the bridge.

7 Two frontiers

▷ **The h^- criterion in the shadow, outside the bridge.** At $q' = 31$ with $n = q'$ the real defect is 3 while the shadow drops by 1: the two are not compared against the same ring. Theorem 25 says which real ring the shadow is to be compared with — not the one of the same n , but the one of the segment recentred at $Nr/2$, whose defect there is also 1 — and that is what dissolves the asymmetry inside the main family. What is open is outside it. When $p \mid N$ — the second source of defect already present in type C — the proof does not stop: $e_1(S)$ stays a unit, the residue ± 1 coming from the sum over all classes and not from N , and the only thing that changes is that the shift becomes $p^{v_p(N)}$. What is open there is narrower, and it is not a step: the fixed ring B^t is A by Proposition 26, the ring of the centred segment $S - Nr/2$, which §2 identifies with a symplectic ring; below depth P that ring follows the ladder (Theorem 27), and what is not settled when $p \mid N$ is its profile beyond it. For the shadow's first layer the question does not arise: Proposition 23 answers it directly, and the answer is $W_1 = 0$. And $p = 2$ is a different problem: q is even, 2^{-1} does not exist modulo q , and even where a centring element exists the proof's recovery of μ from μ^2 and its projectors divide by 2. The statement must be made family by family, as in [14], and not in one piece.

▷ **The stratified reading, and the guess it kills.** The columns are permuted by the strata $X_D = \{c : \gcd(c, q') = D\} \cong (\mathbb{Z}/(q'/D))^\times$, which is the structure of gcd-graphs, whose eigenvalues are Ramanujan sums and which by So's theorem are the only integral circulant graphs [31, 11]. Writing S_D for the restriction of S to X_D , the annihilator splits, $\text{Ann}(S) = \bigcap_D \text{Ann}(S_D)$, so

$$W_1 = \text{codim}_{\mathbb{F}_p[G]} \bigcap_{D \mid q'} \text{Ann}(S_D), \quad G = (\mathbb{Z}/q')^\times,$$

which is the stratified reading in its correct form: when $p \nmid \varphi(q')$ the group algebra is semisimple and a block counts once, as soon as it survives in at least one stratum. Summing the strata instead counts it once per stratum, which is why that overshoots. What does not work is the refinement one wants next. A character of conductor f appears in every stratum with $f \mid q'/D$, so one hopes it either survives in all of them or in none, which would make $\text{codim Ann}(S_D)$ a divisor sum in q'/D and give a count by conductor. It is false: survival depends on the stratum, and in 76 of 416 stable cells even the unit stratum $D = 1$, the largest, fails to see the whole first layer. A witness with a semisimple group algebra is in the shadow, the segment $\{0, \dots, 13\}$ of $\text{GL}(14)$ at $q' = 14$, $p = 5$ ($q = 70$, and $5 \nmid \varphi(14)$), where the layer dimension is $\varphi(14) = 6$, $W_1 = 4$ and $\text{codim Ann}(S_1) = 3$: a block dies on the units and lives on a smaller stratum. So a count by conductor that assumes a block survives uniformly across the strata is false; a classification by conductor that carries the data of the strata is not ruled out, and what is missing is a description of which strata a block survives in.

Remark 29 (a negative result). $\text{gr}_1(A)$ is *not* the Galois orbit of a single element of $\bar{R}$: it is the span of all the coefficients of one polynomial. Three attempts to describe it as such — the module generated by $\sum_c S(c)\omega^c$, the sum of the ranks stratum by stratum, and the module generated by $\sum_c S(c)U_c\omega^c$ — give 181, 169 and 101 of the 193 stable cells, against 193 for the matrix of Theorem 10.

8 What this computes, and a tool that computes it

Two of the statements above replace a computation by a formula, and that is worth saying in the currency of whoever has to do the computation. The object is an order in a cyclotomic field and the question is its conductor; that is the everyday task in the algorithmics of complex multiplication, where the conductor of a CM order is what one computes in order to identify an endomorphism ring or to place a curve in an isogeny graph [10, 2]. In that setting the two read as follows.

A filter that costs nothing. Theorem 16 says that p can divide $[\mathcal{O} : A]$ only when r divides one of $n - 1, n, n + 1$, and Corollary 17 narrows that to $n, n - 1$ for the shadow. That is a divisibility test: no lattice, no matrix, no layers. It certifies maximality at every prime outside the families in the time it takes to reduce n modulo r , and it can be run over as many pairs as one likes before any real work starts.

Half the degree. Theorem 25 gives $v_p([\mathcal{O} : B_n]) = 2v_p([\mathcal{O}_+ : R]) + Pd$. The left side is an invariant of an order of degree $\varphi(q)$; the right side is computed in the real subfield, of degree $\varphi(q)/2$, plus one multiplication. This is the shape of statement that the CM literature has for the case where the real order is *maximal* [10, Lemmas 7, 8]; what is new is that it does not need that.

The tool also implements, as **ceros** and **fila**, the two counts of the companion note [32]: the vanishing components on the diagonal, and the rows a quadratic character occupies — exactly those with $\left(\frac{2}{n}\right) = 1$, which [32] proves by showing that a completely multiplicative sign pattern with zero weighted sum exists at every admissible length.

The tool. `segmento.py` travels with this note: one file, Python 3 standard library only, no installation and no computer algebra system. It exposes **familias**, **perfil**, **indice**, **punte**, **ceros** and **fila** — one routine per statement of this note and of [32], named after it — as a library and as a command line, together with `segmento_SU`, which returns the segment of g_q with its modulus ($M = 2q$ for n even, as in §2). Running `python segmento.py autotest` reproduces, from scratch, the worked examples of this note — the profiles (1, 11, 21, 22) at $q = 207$ and (1, 15, 29, 30) at $q = 279$, the index $v_3([\mathcal{O} : B_{23}(207)]) = 33$ — and those of [32]. The slow path, **indice**, is kept deliberately: it is the control against which the formulas are checked, and the check is part of the test. It is also deposited on its own, under CC0 and with a DOI of its own [15], so that it can be found, used and cited by someone who never reads this note — which is the audience an application has.

Measured: The shortcut against the slow path in 9 cells with $\varphi(q) \leq 132$: the same integer in every cell, and about half the wall clock at the largest of them. The factor is the ratio of the two degrees and not an asymptotic claim — both routes run the same layer code, and the only difference is that one closes in degree $\varphi(q)$ and the other in $\varphi(q)/2$; the times themselves are a property of the machine and the archived run records them. The script that times it, `medir_atajo.py`, travels with the note as well.

What the tool does *not* do is hide the honesty floor of the note: where a statement is measured and not proved — the count of vanishing components at a composite f , where that count needs f prime — the corresponding routine says so and declines; and where the layer profile is cut at depth E before it fills, **indice** and **punte** return a lower bound and say so; and at $q' \leq 2$, outside Theorem 10, **indice** declines. The guards are exactly these three and no others. The shortcut is not guarded at $u \leq d/2$, because Proposition 26 gives $R = A$ unconditionally; and it is not guarded at $p \mid N$, and should not be, because there the number is justified and what stays open is the deeper profile of the real side, not a step.

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Primary 11R18; Secondary 11R29, 11R04, 13H10, 13B22, 05C50, 20G05, 11Y40.

Declaration of competing interest

The author declares no competing interest and no funding.

Data availability

No external data were used. The computational data — every number labelled *measured*, and every figure — are generated by scripts that travel with this note together with their archived output. Among them is `segmento.py` of §8, a single dependency-free file that implements the statements of this note and verifies its own numbers with `python segmento.py autotest`; it is released into the public domain under CC0 so that it can be used, cut up or rewritten without asking, and is deposited as a citable software record of its own [15].

Declaration on generative AI

This note was written with Claude (Anthropic) as an assistant, under the author’s direction. Every statement marked *measured* is produced by a script archived with the note; the author is responsible for the content.

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